

Interventional Nephrology: Managing Dialysis Access and Kidney Biopsy

Ammar Almekhmi | Alian A. Albalas | Brett Cullis

CHAPTER OUTLINE

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KEY POINTS

- Interventional nephrologists are specialists in placing and maintaining vascular and peritoneal dialysis access options for patients with end-stage kidney disease.
- Dialysis access is the lifeline for patients with end-stage kidney disease. Decisions regarding vascular access should be based on the patient's life-plan needs, circumstances, and vessel characteristics.
- Once matured, the arteriovenous fistula is the preferred dialysis access due to its low complication rate and good lifespan as compared with arteriovenous graft and tunneled dialysis catheters.
- Neointimal hyperplasia of the dialysis access is the key pathologic lesion that is associated with access dysfunction and subsequent lifespan compromise.
- Endovascular arteriovenous fistula treatment is the breakthrough in dialysis access creation over the last decade. However, more studies are needed to evaluate the long-term patency outcomes, complications, and cost effectiveness as compared to surgical fistulas.
- Peritoneal dialysis catheters placed at the bedside by nephrologists reduce time to start of dialysis and have good outcomes.

During the lifetime of a hemodialysis patient, he or she may require many vascular access procedures including placement of temporary or tunneled hemodialysis catheters, arteriovenous (AV) fistula and AV graft placements and revisions, surgical or percutaneous thrombectomies of AV fistulas and grafts, and other related endovascular

procedures for AV access. Dialysis access procedures may be performed by vascular surgeons, interventional radiologists, and nephrologists. In recent years, interventional nephrologists have expanded the services they perform to include primary AV fistula and graft creation, endovascular AV fistula creation, insertion of tunneled hemodialysis

and peritoneal dialysis catheters, and performing kidney biopsies. Issues relating to these procedures are discussed in this chapter.

The ideal dialysis access is the one that can provide reliable adequate dialysis with minimal interventions and complications and is consistent with a patient's life plan. Hemodialysis and peritoneal dialysis, as well as the need for dialysis access and the management of access-related complications, are discussed further in [Chapters 62 and 63](#).

INTERVENTIONAL NEPHROLOGY IN MAINTENANCE OF HEMODIALYSIS VASCULAR ACCESS

During the course of their vascular access history, many patients undergo frequent placement of temporary or tunneled hemodialysis catheters, revision of a permanent access, surgical or percutaneous thrombectomies, and other related endovascular procedures. Since the publication of the initial Kidney Disease Outcomes Quality Initiative (KDOQI) guidelines in 1997 and subsequent updates, interventional nephrologists (and interventional radiologists) have become directly involved in providing vascular access procedures. Interventional nephrology was pioneered by Gerald Beathard¹ and subsequently adopted by nephrologists at other medical centers.²⁻⁷ As a consequence, dialysis access procedures are increasingly being performed by appropriately trained nephrologists who know the patients extremely well and are focused on these procedures.⁸

In 2000, a group of interventional nephrologists and radiologists, under the leadership of Dr. Beathard, formed the American Society of Diagnostic and Interventional Nephrology (ASDIN).⁹ This organization provides certification to interventional nephrologists, as well as accreditation to the institutions involved in the practice and teaching of interventional procedures in the nephrology specialty.⁵ With time, the profile of endovascular procedures performed at freestanding vascular access centers has evolved to perform all dialysis access procedures including the surgical and endovascular AV fistula (EndoAVF) creation.^{10,11} A comprehensive list of procedures typically performed by interventional nephrologists is provided in [Table 67.1](#).

It is worth mentioning that when interventional nephrologists perform the access procedures, hospitalizations related to vascular access-related complications and missed outpatient hemodialysis sessions because of vascular access problems decrease.^{7,12-14} Some centers employ a multidisciplinary model consisting of a joint interventional radiology/nephrology program, with interventional nephrologists and interventional radiologists sharing the same radiology suites, and working side by side to perform all dialysis access procedures, profiting from the existing technical, clinical, imaging, and surgical competencies at the same institution.¹⁵

A key element of any successful interventional program, whether it involves radiologists or nephrologists, is active collaboration with nephrology and dialysis programs to track outcomes of the procedures and implement timely quality improvement initiatives. This can be best accomplished by

Table 67.1 Procedures Performed by Interventional Nephrologists

Placement of nontunneled and tunneled dialysis catheters
Exchange and removal of tunneled dialysis catheters
Preoperative vascular mapping
Coiling and ligating accessory veins
Creating AV fistula endovascularly (EndoAVF)
Diagnostic angiograms and angioplasty of grafts and fistulas
Deployment of endoluminal stents for peripheral and central stenosis
Thrombectomy of grafts and fistulas
Surgical and endovascular AVF creation
Salvage procedures for immature fistulas
Placement of peritoneal dialysis catheters
Diagnostic renal ultrasound
Percutaneous renal biopsy

having dedicated vascular access coordinators who maintain prospective, computerized records of all procedures performed and follow patients up regularly.¹⁵

Vascular access procedures and their subsequent complications represent a major cause of anxiety, morbidity, hospitalization, and cost for patients on maintenance hemodialysis.¹⁶⁻²⁰ In an early publication from 1996, more than 20% of hospitalizations in U.S. hemodialysis patients were vascular access related, and the annual cost of vascular access morbidity is close to 1 billion U.S. dollars.¹⁹ Thamer and colleagues²¹ reported that Medicare paid 2.8 billion U.S. dollars for dialysis vascular access services in the United States. Complications that occur commonly with AV fistulas and grafts are stenosis and thrombosis, which contribute to the rising costs of vascular access care and patient morbidity. AV grafts are prone to recurrent stenosis and thrombosis and often require multiple interventions to ensure long-term patency. On the other hand, AV fistulas have a much lower incidence of stenosis and thrombosis than AV grafts that require fewer interventions.^{19,20,22}

OVERVIEW OF VASCULAR ACCESS FOR DIALYSIS

Patients with end-stage kidney disease (ESKD) require a reliable dialysis access to optimize their survival, minimize medical complications, and enhance quality of life.²³ The ideal dialysis access is the one that fulfills the patient needs, is easy to place and to use, delivers sufficient blood flows indefinitely, and is free of complications. Preparation for dialysis access should start early in patients with progressive decline of kidney function or with an estimated glomerular filtration rate (eGFR) 15 to 20 mL/min/1.73 m².²⁴ The preferred dialysis access is mainly dependent on the patient's needs and preferences, as well as dialysis access eligibility. Accordingly, while a young, active predialysis patient can initiate dialysis using peritoneal dialysis, an elderly patient with multiple comorbidities who needs urgent dialysis can start with a central venous catheter (CVC). Dialysis access includes AV grafts and fistulas, CVC, and peritoneal catheter (PD).

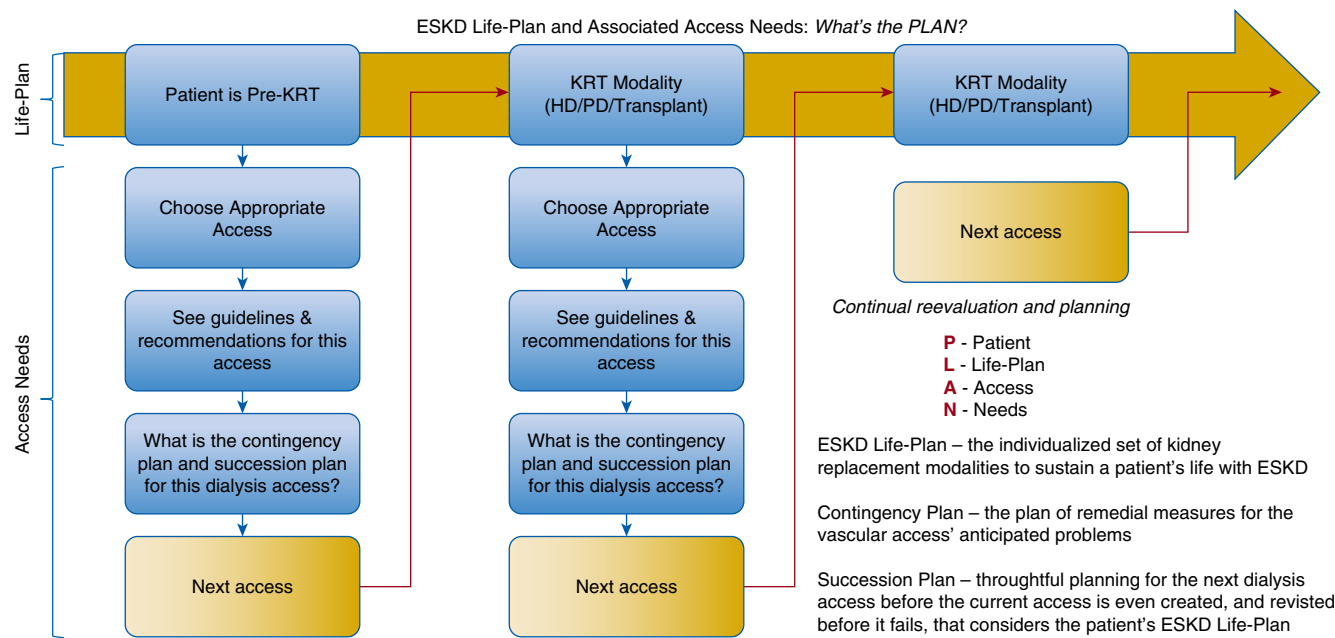


Fig. 67.1 End-Stage Kidney Disease and Dialysis Access Life-Plan: *What's the P-L-A-N?* HD, Hemodialysis; KRT, kidney replacement therapy; PD, peritoneal dialysis. (Adapted from Lok CE, Huber TS, Lee T, et al; National Kidney Foundation. KDOQI Clinical Practice Guideline for Vascular Access: 2019 Update. *Am J Kidney Dis.* 2020;75(suppl 4):S1–S164.)

A reliable and durable vascular access, via either a central venous catheter, AV fistula, or graft is a critical requirement for providing adequate hemodialysis. In 2019, the KDOQI Vascular Access Guidelines introduced the ESKD Life-Plan, which is an individualized and comprehensive map for dialysis modalities and vascular access (Fig. 67.1).²⁴

ARTERIOVENOUS FISTULAS AND ARTERIOVENOUS GRAFTS

Native AV fistulas are created by connecting a high-pressure artery to a low-pressure vein, leading to vein dilatation and arterIALIZATION and subsequent increased blood flow. To be used reproducibly for dialysis, an AV fistula must have a large enough diameter to be safely cannulated with large-bore dialysis needles, must be superficial enough to be cannulated, and have a sufficiently high-access blood flow to permit blood flow ≥ 350 mL/min. Increases in blood flow and draining vein diameter occur fairly rapidly following AV fistula creation. Whereas blood flow in a normal radial artery is only 20 to 40 mL/min, it increases more than 10-fold within a few weeks of fistula creation. In one study, the mean-access blood flow in successful AV fistulas was 634 mL/min 2 weeks postoperatively²⁵; in a second study it was 650 mL/min 12 weeks after AV fistula creation.²⁶ The mean-access blood flows and AV fistula diameters do not change significantly over the second, third, and fourth months following AV fistula creation.²⁷ As such, the suitability for cannulation of AV fistula (maturation) can be determined within 4 to 6 weeks of fistula creation.

AV graft is an artificial connection, commonly made of polytetrafluoroethylene (PTFE), between an artery and a vein. Blood flow is immediately high in an AV graft; therefore these can be cannulated early, ideally within 3 to 4 weeks post surgical creation once the swelling around the graft has subsided.

Dialysis catheters include temporary large-bore venous catheters, which are usually only used short term, and tunneled cuffed permanent CVCs. Catheters may be the first choice of access for some patients, may be needed as a bridge in patients waiting for an AV fistula to mature or an AV graft to heal, or may serve as an access of last resort in patients who have exhausted all AV fistula or graft options.^{24,28,29} As compared with patients who continue to dialyze with catheters, those who switch from a catheter to an AV fistula or graft have a substantially lower mortality,^{30,31} as well as improvements in serum albumin and erythropoietin responsiveness.³²

The KDOQI 2006 guidelines recommended a target use of AV fistulas in at least 40% of hemodialysis patients and catheter use in only 10% of prevalent patients.³³ These guidelines in combination with the Fistula First Initiative resulted in increased AV fistula use between 2003 and 2012 in the United States. However, the KDOQI 2019 guidelines suggest that the dialysis access choice should be more patient centered, reflecting the ESKD Life-Plan needs, based on the patient circumstances and vessel characteristics.²⁴ To meet these recommendations of the new guidelines, a multidisciplinary approach is needed, which includes the patient, nephrologists, vascular surgeons, interventionalists and social workers, and engages with the individual patient for optimal dialysis access planning.³⁴ For patients who choose hemodialysis, a vascular access plan becomes an important aspect of care, ideally completed well before hemodialysis initiation.

SELECTION OF VASCULAR ACCESS LOCATION

The planning and preservation of vascular sites for dialysis access is crucial in patients with ESKD. Prior injury and scarring in the relevant vessels may lead to AVF maturity failure, recurrent AVG thrombosis, and CVC-related

complications. Therefore the preservation of peripheral and central veins and arteries from any damage as much as possible is highly recommended. Whenever possible, venipunctures and peripherally inserted catheters should be avoided if possible among patients with advanced CKD (see www.saveyourvein.org).²⁴

If necessary, venipunctures and blood draws should be performed in the hand of one arm only, and the nondominant arm should generally be protected from any punctures as this is the preferred side for the AV access unless there are reasons to choose the other side (e.g., known subclavian stenosis). In patients with anticipated prolonged duration on dialysis, a distal-to-proximal approach should be considered to preserve more proximal vessels for future access. This approach begins with forearm AVF creation (or forearm AVG creation if vessels are deemed unsuitable for AVF) and moving more proximally as needed to proximal forearm AVF creation using perforator vein (EndoAVF), upper extremity AVF creation (brachiocephalic AVF), or brachiocephalic AVF versus upper-extremity AVG creation based on vessel characteristics and operator discretion.²⁴ On the other hand, patients who require dialysis urgently without previous access planning are expected to have limited time of dialysis (<1 year), and the dialysis therapy can be performed via CVC. Finally, for those patients who have no prior access planning, expected to be on dialysis for prolonged time (>1 year), dialysis therapy can be initiated via a CVC to bridge to AVF maturation or an upper- (or lower)-extremity loop AVG with early cannulation based on patient needs and operator experience.²⁴ Peritoneal dialysis may also be an option for some patients.

PREOPERATIVE VASCULAR MAPPING

In clinical practice, a detailed history, physical examination, and duplex ultrasonography are commonly used in preparation for AV access creation. Detailed physical examination should assess the peripheral pulse and blood pressure in both extremities, examine the arm for edema, evaluate signs for heart failure, and identify marks of old lines and catheters, previous surgeries, and cardiac devices.

Preoperative vein mapping using vessel duplex ultrasonography was recommended in previous KDOQI 2006 guidelines solely on the basis of expert opinion to assess

vein anatomy and diameter. However, several disadvantages are associated with duplex ultrasonography including operator errors, cost, patient inconvenience, and effect of temperature on vessel size. As such, KDOQI 2019 suggests the use of preoperative ultrasonography in a selected population, such as the elderly, females, those with multiple comorbidities (coronary and peripheral arterial disease), those with previous peripheral or central lines, and morbidly obese patients. Venography can be used when central venous stenosis is suspected.²⁴ As discussed later, however, vascular ultrasonography can impact decision making and the success of AV access creation and is recommended if the operators are skilled in this technique.

Allon and colleagues³⁵ have suggested a minimal vein size of >2.5 mm for AVF creation and >4 mm for AVG creation and an artery size of >2 mm. When these thresholds were used, the proportion of patients having an AV fistula placed increased from 34% to 64% as compared with historical controls, and the proportion of patients dialyzing with an AV fistula doubled from 16% to 34%.³⁵

Dageforde and colleagues³⁶ retrospectively evaluated 158 patients who had brachiocephalic or brachiocephalic AVF and found that a vein size of >3.4 mm had a higher AVF maturation rate compared with those with vein size of <3.4 mm.³⁶

Most surgeons obtain intraoperative vascular ultrasonography. A prospective pilot study compared the surgeon's decision about access placement in 70 consecutive patients with CKD, before and after the results of preoperative vascular mapping were provided to the surgeon.³⁷ In almost one third of the patients, the surgeon changed his or her mind about the intended access procedure after receiving the mapping results. In most of these cases, the surgeon decided to place an AV fistula rather than an AV graft or used a different location for the fistula creation.³⁷ Similar increases in AV fistula placement following the introduction of preoperative vascular mapping have been documented by other investigators (Table 67.2),³⁵ although a reduction in primary AV fistula failure has not been a consistent finding in all studies.^{35,38-44} Two randomized clinical trials have compared outcomes of new AV fistulas placed after preoperative ultrasound vascular mapping, as compared with those placed after clinical evaluation alone. In a Turkish study of 70 patients, the two treatment groups had similar primary and secondary AV fistula

Table 67.2 Effect of Preoperative Vascular Mapping on Vascular Access Outcomes

Reference	% Fistulas Placed		% Primary Fistula Failure		% Prevalence of Fistula Use	
	Pre-VM	Post-VM	Pre-VM	Post-VM	Pre-VM	Post-VM
Silva, 1998 ⁴⁰	14	63	36	8	8	64
Ascher, 2000 ³⁸	0	100	N/A	18	5	68
Gibson, 2001 ³⁹	11	95	18	25	N/A	N/A
Allon, 2001 ³⁵	34	64	54	46	16	34
Sedlacek, 2001 ⁴²	N/A	62	N/A	25	N/A	N/A
Mihmanli, 2001 ⁴³			25	6		
Miller, 1997 ⁴⁴	N/A	76				
Kakkos, 2011 ⁴¹	12	53	32	18		

N/A, Not available; VM, preoperative vascular mapping.

Adapted from Allon M, Robbin ML. Increasing arteriovenous fistulas in hemodialysis patients: problems and solutions. *Kidney Int*.

survival.⁴⁵ In a British study of 218 patients, the cohort that underwent a preoperative ultrasound had a lower incidence of immediate AV fistula failure (4% vs. 11%, $P = 0.03$).⁴⁶ Primary (intervention-free) AV fistula survival was not different between the two groups (65% vs. 56%, $P = 0.08$),⁴⁶ but the assisted primary (thrombosis-free) AV fistula survival was superior in the group undergoing preoperative ultrasound mapping (80% vs. 65%, $P = 0.01$).

PATHOGENESIS OF VASCULAR ACCESS STENOSIS IN AV FISTULAS AND AV GRAFTS

During access creation process, a significant vascular injury is encountered due to vessel manipulation, clamping, and hypoxia leading to local cytokine release and inflammation.⁴⁷ After making the connection between the artery and vein (or graft material in AVG), the flow disturbances and increased shear pressure at the anastomosis can aggravate the vascular injury, leading to inward remodeling.⁴⁸ As mentioned earlier, the multicenter Hemodialysis Fistula Maturation (HFM) study revealed a progressive increase in AV fistula diameter and blood flow in the 6-week period following AV fistula creation.⁴⁹ Incorporating data from the Dialysis Access Consortium (DAC) Fistula Trial, Dember and colleagues⁵⁰ reported that AV fistula nonmaturation rates in the United States are approximately 60% even among centers of excellence. The magnitudes of postoperative AV fistula diameter and blood flow were directly associated with the preoperative arterial function, assessed by nitroglycerin-mediated dilation and flow-mediated dilation.⁵¹ Studies from experimental models have provided valuable insight into the downstream biology of AV fistula nonmaturation.⁴⁸ These studies have shown that proper balance of expression of protective factors that regulate oxidative stress and

endothelial function—such as heme-oxygenase-1 (HO-1) and nitric oxide synthase (eNOS) and inhibition of chemokines and mediators that modulate inflammation and vessel wall remodeling, such as monocyte chemoattractant protein-1 and matrix metalloproteinases (MMP-2 and MMP-9)—are crucial in increasing AV fistula blood flow and vasodilation and inhibiting venous neointimal hyperplasia and stenosis after AV fistula creation.⁵²⁻⁵⁴ From clinical studies, gene polymorphisms in HO-1⁵⁵ and factor V Leiden⁵⁶ have been associated with AV fistula failure.

The preexisting health of the vein and artery at the time of access creation and its impact on AV graft and fistula outcomes are important.⁵⁷ Several studies have shown that preexisting venous neointimal⁵⁸⁻⁶¹ and arterial neointimal hyperplasia⁶² are present in the large majority of vessels collected at the time of new vascular access surgery. Preexisting arterial intimal hyperplasia and arterial microcalcification has been associated with early AV fistula failure.⁶² Preexisting venous neointimal hyperplasia has not been associated with early postoperative AV fistula stenosis.⁶³ Similarly, a single-center study of patients undergoing a two-stage AV fistula found no significant association between preoperative venous intimal hyperplasia and postoperative neointimal hyperplasia or clinical AV fistula survival.⁶⁴ Analysis of the HFM study observed only a weak association between the magnitude of preoperative venous intimal hyperplasia and the postoperative AV fistula blood flow, stenosis, and clinical maturation.⁶⁵ The data suggest that outward remodeling (vasodilation) after AV fistula creation may play a more important role in the maturation process than preoperative venous intimal hyperplasia.

In AV fistulas, neointimal hyperplasia is the most common histologic change seen in both early and late fistula failures^{48,66-70} (Fig. 67.2). The predominant cellular phenotypes present within the neointima are myofibroblasts.⁶⁶⁻⁷⁰ In AV fistula

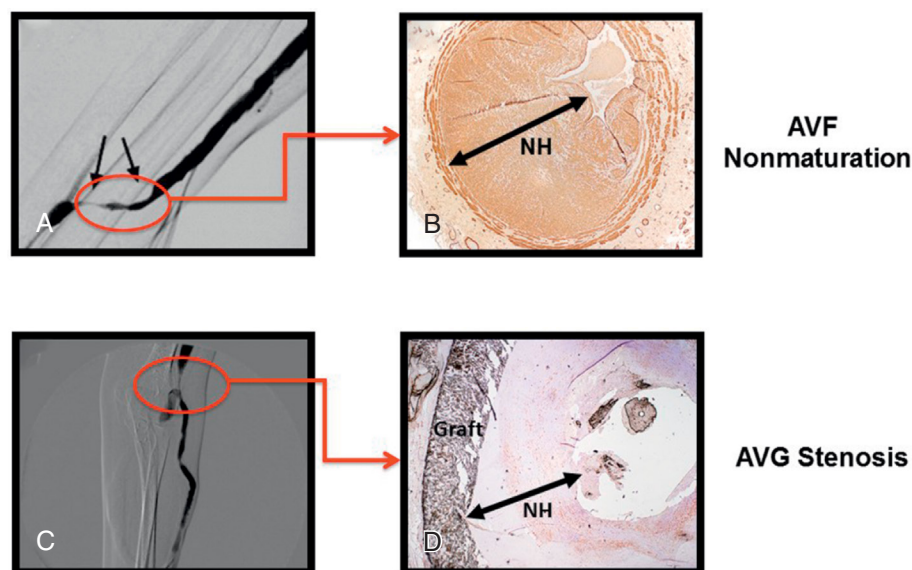


Fig. 67.2 Histologic and angiographic lesions of venous stenosis in arteriovenous fistula (AVF) and arteriovenous graft (AVG). (A) Angiographic and (B) histologic features of AVF nonmaturation. Note aggressive venous neointimal hyperplasia (NH) at vein-artery anastomosis. (C) Angiographic and (D) histologic features of AVG stenosis. Note aggressive neointimal hyperplasia at graft-vein anastomosis. Arrows show histologic features at the site of the angiographic venous stenosis in both AVF and AVG. Images are α -smooth muscle actin stain and magnification is 4 $\times$. (Adapted and reprinted from Lee T, Misra S. New insights into dialysis vascular access: molecular targets in arteriovenous fistula and arteriovenous graft failure and their potential to improve vascular access outcomes. *Clin J Am Soc Nephrol*. 2016;11(8):1504–1512, with permission from the American Society of Nephrology.)

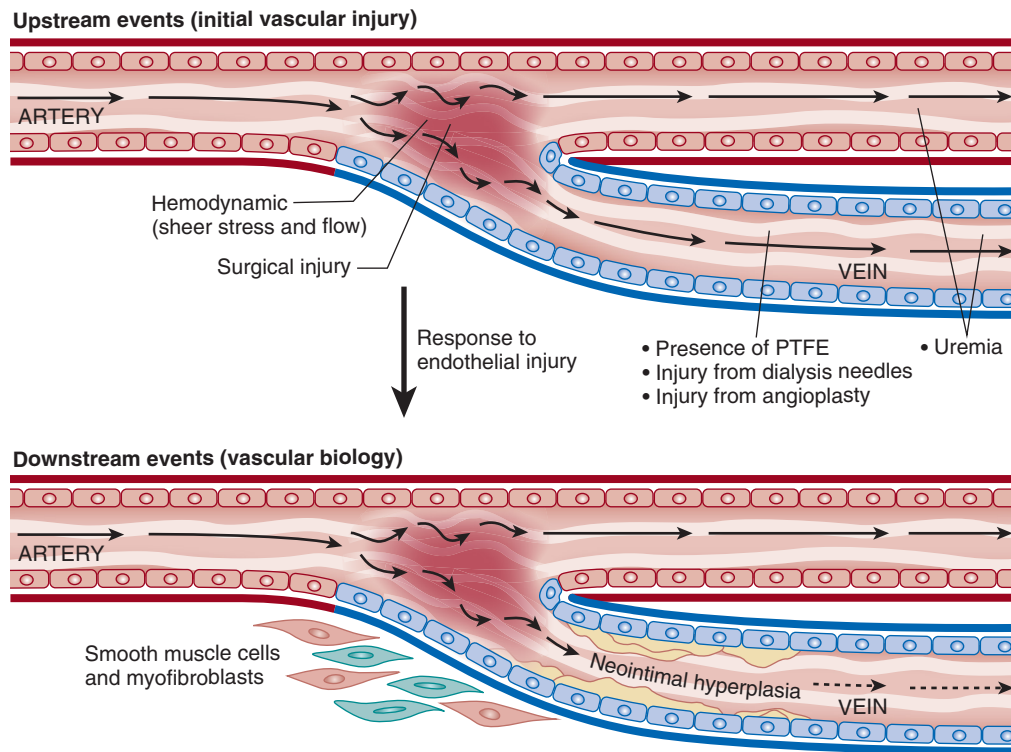


Fig. 67.3 Upstream and downstream events in hemodialysis vascular access dysfunction. Upstream events result in initial vascular injury. Downstream events are the vascular biologic response to upstream injury. Downstream biology involves mediators of oxidative stress and inflammation that regulate activation, proliferation, and migration of fibroblasts; smooth muscle cells; and myofibroblasts. *PTFE*, polytetrafluoroethylene. (Adapted and reprinted from Lee T. Novel paradigms for dialysis vascular access: downstream vascular biology—is there a final common pathway? *Clin J Am Soc Nephrol*. 2013;8(12):2194–2201, with permission from the American Society of Nephrology.)

nonmaturation, inadequate vasodilation (poor outward remodeling), in addition to neointimal hyperplasia, also plays an important role.^{68,71,72} The most common mechanism of vascular access dysfunction in AV grafts is the development of venous neointimal hyperplasia at the vein-graft anastomosis^{48,68,73} (see Fig. 67.2). Venous neointimal hyperplasia is characterized by myofibroblasts and extracellular matrix components within the neointima.⁷³ In addition, angiogenesis within the adventitia and a macrophage layer lining the perigraft region may also develop.⁷³

The pathogenesis of neointimal hyperplasia after AV fistula creation or graft placement involves a cascade of events that are best divided into upstream and downstream events⁷¹ (Fig. 67.3). Upstream events are characterized as the initial injuries to the vascular endothelium and smooth muscle cells, which include 1. surgical trauma at the time of AV access creation; 2. hemodynamic shear stress at the vein-artery or vein-graft anastomosis; 3. inflammatory response to the PTFE grafts; 4. injury from vessel cannulation; 5. the effects of uremia on the veins and arteries leading to endothelial dysfunction; and 6. angioplasty-related injury.^{68,69,71} Downstream events represent the vascular biological response to these upstream vascular injuries (see Fig. 67.3). An inappropriate downstream response will lead to activation, proliferation, and migration of fibroblasts, smooth muscle cells, and myofibroblasts, ultimately leading to the development of neointimal hyperplasia.^{68,69,71,72} In AV fistulas, vasodilation is a necessary downstream response for maturation, in addition to inhibition of neointimal hyperplasia^{48,71,72} (Fig. 67.4).

CLINICAL MONITORING OF ARTERIOVENOUS DIALYSIS ACCESS

According to KDOQI 2019 guidelines, physical examination is considered a standard, accepted, cost-effective method for monitoring dialysis access function. Before a dialysis session, a “1-Minute Access Check” physical examination of the access by an experienced person carries a high sensitivity and specificity (<https://esrdncc.org/en/patients/ffcl-for-patients/lifeline-for-a-lifetime/step-eight-the-one-minute-access-check/>). This includes inspection of the arm, chest, neck and face; old marks of CVC and other lines; palpation of the entire access; the arm elevation test; and the pulse and augmentation tests.²⁴ Clinical findings during a physical examination usually suggest the clinical diagnosis in most cases. For instance, the presence of ipsilateral extremity edema raises the suspicion of central venous stenosis. On the other hand, failure of an AVF to collapse upon elevating the extremity indicates a venous outflow lesion. The clinical findings during dialysis can be complementary to physical examination findings. For example, the presence of high venous pressures, recirculation, or prolonged bleeding after removing the needles may point toward an outflow stenosis.

ARTERIOVENOUS GRAFT MALFUNCTION

SURVEILLANCE FOR GRAFT STENOSIS

About 80% of AV graft failures are due to thrombosis, whereas 20% are due to infection.^{74,75} Thus improving AV

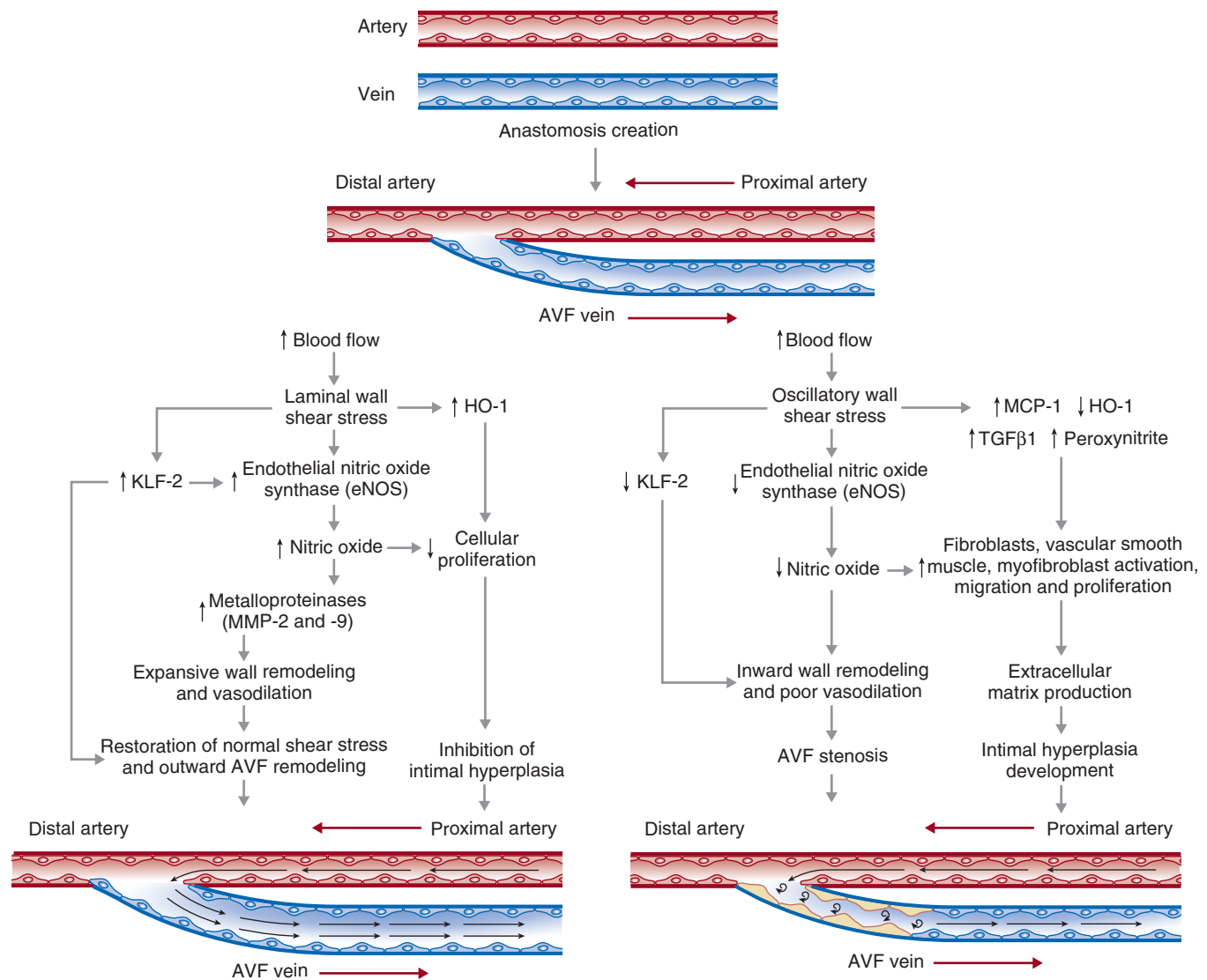


Fig. 67.4 Pathophysiologic events of successful arteriovenous fistula (AVF) maturation and AVF maturation failure. Left panel describes events of successful AVF maturation and right panel describes events of AVF maturation failure. Successful AVF maturation is dependent on outward vascular remodeling and inhibition of neointimal hyperplasia, regulated through nitric oxide production and appropriate regulation of matrix metalloproteinases. Fibroblast, smooth muscle cell, and myofibroblast activation, migration, and proliferation play a key role in neointimal hyperplasia development and AVF maturation failure. Mediators such as heme-oxygenase-1 (HO-1), monocyte chemoattractant protein (MCP-1), kruppel-like factor-2 (KLF-2), transforming growth factor β (TGF β 1), and high levels of local oxidant stress (e.g., peroxynitrite) play essential roles in regulating cellular proliferation and neointimal hyperplasia development. (Adapted and reprinted from Lee T, Misra S. New insights into dialysis vascular access: molecular targets in arteriovenous fistula and arteriovenous graft failure and their potential to improve vascular access outcomes. *Clin J Am Soc Nephrol.* 2016;11(8):1504–1512, with permission from the American Society of Nephrology.)

graft longevity requires implementing measures to reduce the frequency of AV graft thrombosis. Reports from the U.S. Renal Data System showed that the number of angioplasties and thrombectomies (“declots”) for AV grafts had substantially increased from 0.49 to 1.10 (rate per patient year) and 0.15 to 0.48, respectively, from 1998 to 2007.²³ Among AV grafts referred for thrombectomy, a significant proportion have underlying stenosis, most commonly at the venous anastomosis, recipient/draining vein, or central veins.⁷⁶⁻⁷⁸ This observation suggests that prophylactic angioplasty of hemodynamically significant AV graft stenosis may reduce the frequency of AV graft thrombosis and thereby increase cumulative AV graft survival. An early study suggested that

measurement of dynamic venous pressures in AV grafts during consecutive hemodialysis sessions under carefully standardized conditions may be useful.⁷⁹ Persistent elevation in the venous pressure measured at a low dialysis blood flow (200 mL/min) was associated with hemodynamically significant stenosis, and referral for prophylactic angioplasty in such cases led to a reduced frequency of AV graft thrombosis from approximately 0.6 to 0.2 events per year. This finding stimulated a large volume of subsequent studies aiming to answer two fundamental questions: 1. identifying noninvasive methods to screen for AV graft stenosis and 2. evaluating whether stenosis surveillance and prophylactic angioplasty improved AV graft outcomes.

Table 67.3 Methods of Stenosis Monitoring**Clinical Monitoring**

Physical examination (abnormal bruit, absent thrill, distal edema)
 Dialysis abnormalities (prolonged bleeding from needle sites, difficult cannulation)
 Unexplained decrease in Kt/V or urea reduction ratio

Surveillance

Static dialysis venous pressure (adjusted for systemic pressure)
 Access blood flow
 $Q_a < 600$ mL/min
 Q_a decreased by >25% from baseline
 Doppler ultrasound

Kt/V, Volume-indexed clearance-time product.

Table 67.4 Positive Predictive Value of Monitoring Methods for Graft Stenosis

Surveillance Method	No. of Measurements	Positive Predictive Value
Clinical Monitoring		
Cayco, 1998 ⁸¹	68	93%
Robbin, 1998 ⁸²	38	89%
Safa, 1996 ⁸⁰	106	92%
Maya, 2004 ⁷⁶	358	69%
Robbin, 2006 ⁸³	151	70%
Static Venous Pressure		
Besarab, 1995 ⁸⁴	87	92%
Flow Monitoring		
Schwab, 2001 ⁸⁵	35	100%
Moist, 2003 ⁸⁶	53	87%
Ultrasound		

A variety of methods have been used for detection of hemodynamically significant AV graft stenosis (Table 67.3). Clinical monitoring consists of using information that is readily available from physical examination of the AV graft, abnormalities experienced during the dialysis sessions (difficult cannulation or prolonged bleeding from the needle puncture sites), or unexplained decreases in the dose of dialysis, urea reduction ratio, or volume-indexed clearance-time product (Kt/V).^{76,77,80} The positive predictive value of clinical monitoring has been reported to be between 69% and 93% (Table 67.4).^{76,80,83} AV graft surveillance uses noninvasive tests requiring specialized equipment or technician training that is not obtained as part of the routine dialysis treatment. These include measurement of static dialysis venous pressure (normalized for the systemic pressure), measurement of the access blood flow, or duplex ultrasound to directly evaluate stenosis. Each

of these monitoring or surveillance tools has been reported to have a positive predictive value for AV graft stenosis of between 70% and 100% (see Table 67.4).⁸³⁻⁸⁶ The negative predictive value has not been studied systematically, as it would require obtaining routine angiography in patients whose screening test is negative. However, the negative predictive value can be inferred from the proportion of AV graft thromboses not preceded by abnormalities of AV graft surveillance, which is around 25%.⁸⁷

In contrast, the predictive value of surveillance methods for AV graft thrombosis is much less impressive. Thus when AV grafts with abnormal monitoring criteria suggestive of stenosis are observed without preemptive angioplasty, only about 40% of the AV grafts clot over the following 3 months.^{88,89} In practice, this means that in any program of AV graft monitoring, about half of preemptive angioplasties performed may be unnecessary. Unfortunately, there are no reliable tests to distinguish the subset of AV grafts with stenosis that will progress to thrombosis from those that will remain patent without any intervention.

Several observational studies have documented that introduction of a monitoring or surveillance program for AV graft stenosis with preemptive angioplasty lowered the frequency of AV graft thrombosis by 40% to 80%, as compared with the historical control period during which there was no monitoring program (Table 67.5).^{15,79,81,84,90} The promising findings from these observational studies led to the implementation of programs of graft surveillance and preemptive angioplasty aiming to decrease the frequency of AV graft thrombosis.⁹¹ However, over the past few years, the value of AV graft stenosis surveillance has been subjected to rigorous testing in randomized clinical trials (Table 67.6).^{83,86,92-95} Only one of the six randomized trials has demonstrated a benefit of ultrasound AV graft surveillance⁹⁵; the other five showed no benefit of surveillance, despite a substantial increase in the frequency of preemptive angioplasty in the surveillance group.^{83,86,92-94} For example, one study randomized patients with AV grafts to standard clinical monitoring alone or to a combination of clinical monitoring and ultrasound surveillance for stenosis. The patients in the ultrasound group had a 66% higher frequency of undergoing preemptive angioplasty, yet there was no difference between the two randomized groups in terms of frequency of AV graft thrombosis, time to first thrombosis, or likelihood of AV graft failure (Fig. 67.5A and 67.5B).⁸³ Further, a meta-analysis of the randomized studies suggested that the benefit of surveillance with preemptive angioplasty, if present, is likely to be small, with a reduction of AV graft thrombosis not exceeding 23% and a reduction in AV graft failure no greater than 17%.⁹⁶ Therefore a large-scale, multicenter study trial, which takes into account multiple factors involved in access thrombosis, would be required to provide a definitive answer to this controversial question. Accordingly, KDOQI 2019 does not recommend routine AVG surveillance using blood flow measurements, pressure monitoring, or access imaging due to the low quality of evidence.²⁴

The question remains: If the underlying AV graft stenosis is an important predictor of AV graft thrombosis, why is preemptive angioplasty not more successful in reducing AV graft thrombosis? The fundamental problem appears to be the short-lived efficacy of angioplasty to relieve AV

Table 67.5 Effect of Surveillance on Graft Thrombosis: Observational Studies

Reference	Surveillance Method	Thrombosis Rate (Per Graft-Years)		
		Historical Control	Surveillance Period	Percent Reduction
Schwab, 1989 ⁷⁹	Dynamic dialysis venous pressure	0.61	0.20	67
Besarab, 1995 ⁸⁴	Static dialysis venous pressure	0.50	0.28	64
Safa, 1996 ⁸⁰	Clinical monitoring	0.48	0.17	64
Allon, 1998 ¹⁵	Clinical monitoring	0.70	0.28	60
Cayco, 1998 ⁸¹	Clinical monitoring	0.49	0.29	41

Table 67.6 Randomized Clinical Trials of Graft Surveillance

Reference	Surveillance Method	No. of Subjects		PTA/yr		Thrombosis-Free Survival at 1 yr		Cumulative Survival at 1 Year	
		Con	Surv	Con	Surv	Con	Surv	Con	Surv
Lumsden, 1997 ⁹²	Doppler US	32	32	0	1.5	0.51	0.47	N/A	N/A
Ram, 2003 ⁹³	Access flow	34	32	0.22	0.34	0.45	0.52	0.72	0.80
	Doppler US		35		0.65		0.70		0.80
Moist, 2003 ⁸⁶	Access flow	53	59	0.61	0.93	0.74	0.67	0.83	0.83
Dember, 2004 ⁹⁴	Static DVP	32	32	0.04	2.1	N/A	N/A	0.74	0.56
Malik, 2005 ⁹⁵	Doppler US	92	97	N/A	N/A	N/A	N/A	0.73	0.93
Robbin, 2006 ⁸³	Doppler US	61	65	0.64	1.06	0.57	0.63	0.83	0.85

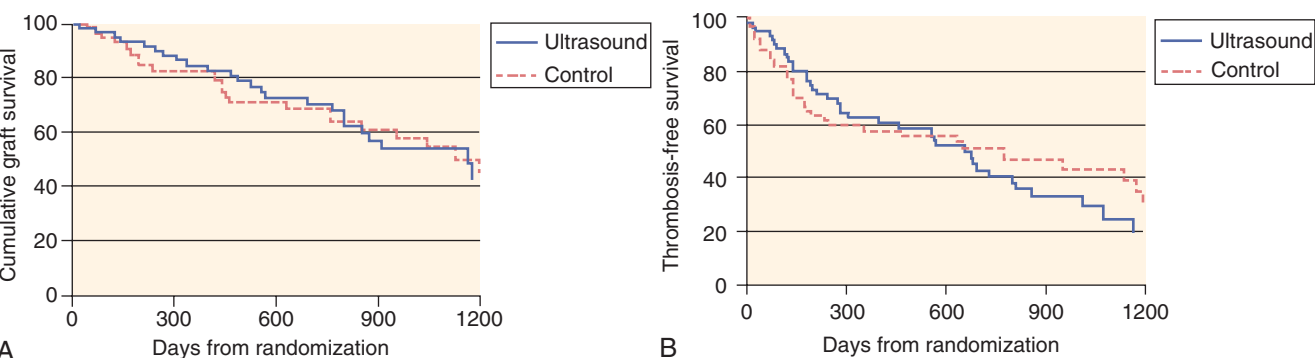


Fig. 67.5 (A) Comparison of cumulative graft survival between randomized patients with clinical monitoring versus clinical monitoring plus regular ultrasound surveillance of grafts. $P = 0.93$ by the log rank test. (B) Comparison of thrombosis-free graft survival between randomized patients with clinical monitoring versus clinical monitoring plus regular ultrasound surveillance of grafts. $P = 0.33$ by the log rank test. (A and B, Reproduced from Robbin ML, Oser RF, Lee JY, et al. Randomized comparison of ultrasound surveillance and clinical monitoring on arteriovenous graft outcomes. *Kidney Int.* 2006;69[4]:730–735.)

graft stenosis. When serial access blood flows have been used as a surrogate marker of successful angioplasty, 20% of AV grafts develop recurrent stenosis within 1 week of angioplasty and 40% within 1 month of angioplasty.^{85,86} In another study, the mean vascular access blood flow

following angioplasty increased from 596 to 922 mL/min and then decreased to 672 mL/min within 3 months after angioplasty.⁹⁷ In addition, the angioplasty-induced injury may actually accelerate neointimal hyperplasia, thereby resulting in recurrent stenosis.⁹⁸ Not surprisingly, patients

Table 67.7 Location of Stenosis in Patients With Grafts Undergoing Angioplasty

Reference	Percent of All Stenotic Lesions				
	VA	VO	CV	IG	AA
Beathard, 1992 ⁷⁸	42	34	4	20	0
Lilly, 2001 ⁷⁷ Elective PTA	55	22	15	6	2.5
Lilly, 2001 ⁷⁷ Thrombectomy	60	14	9	10	7
Maya, 2004 ⁷⁶	62	16	8	12	1.5

AA, Arterial anastomosis; CV, central vein; IG, intragraft; PTA, percutaneous transluminal angioplasty; VA, venous

Table 67.8 Primary Graft Patency After Elective Angioplasty

Reference	No. of Procedures	Primary Patency at:		
		3 Months	6 Months	12 Months
Beathard, 1992 ⁷⁸	536	79	61	38
Kanterman, 1995 ¹⁰¹	90		63	41
Safa, 1996 ⁸⁰	90	70	47	16
Turmel-Rodrigues, 2000 ¹⁰²	98	85	53	29
Lilly, 2001 ⁷⁷	330	71	51	28

undergoing angioplasty for AV graft stenosis require frequent re-interventions due to recurrent stenosis. The median intervention-free patency following AV graft stenosis is only about 6 months.^{76,77} Finally, AV graft surveillance is likely to be most useful in new AV grafts (within 3 months of the creation).⁹⁹ A large proportion of AV grafts fail before there has been adequate opportunity for surveillance to be performed and for hemodynamically significant stenosis to be detected and treated.¹⁰⁰

ANGIOPLASTY OF ARTERIOVENOUS GRAFT STENOSIS

The goal of elective angioplasty is to correct the stenotic lesion that impairs optimal delivery of dialysis, and hopefully delay AV graft thrombosis. Measurements of the degree of stenosis are not always feasible, nor are they entirely objective. Caliper measurement or “eyeballing” are to various degrees, subjective. Even electronic quantitative analysis is not reliable in detecting the various densities and edges of the normal vessels and requires operator adjustment of detected vascular edges (a subjective intervention). Arteriovenous graft loops originating from the brachial artery and those that have anastomosis at a 90-degree angle with the basilic vein are hardest to image in profile, which is necessary in order to get an accurate measurement of stenosis. In practice, the degree of stenosis is estimated subjectively by evaluating the angiogram images before and after angioplasty. In hospital-based centers, intra-graft pressure measurement before and after angioplasty is usually obtained.

The most common location of the stenosis localized by angiography is the venous anastomosis, followed by the peripheral draining vein, the central veins, and intragraft (Table 67.7).⁷⁶⁻⁷⁸ Inflow (arterial anastomosis) stenosis has been rare (<5%) in most series. A number of published series have documented the short-lived primary patency (time to next radiologic or surgical intervention) of AV grafts following elective angioplasty (Table 67.8),^{76-78,80,101,102} with only 50% to 60% patency at 6 months and 30% to 40% at 1 year. This means that, on average, each AV graft requires two angioplasties per year. The primary patency is shorter after angioplasty of central vein stenosis, as compared with other stenotic locations. In one study, the primary patency at 6 months was only 29% for central vein stenosis, as compared with 67% for stenosis at the venous anastomosis.⁷⁸ Most studies have documented

progressively shorter patency after each consecutive angioplasty. Notably, the primary patency of AV grafts following elective angioplasty is not affected by patient age, race/ethnicity, diabetes, or peripheral vascular diseases.⁷⁶ However, the patency tends to be shorter in women than in men.⁷⁶ The primary patency after angioplasty is also not influenced by the location of the AV graft or the number of concurrent stenotic lesions found.⁷⁷ The technical success of an angioplasty procedure may be assessed in several ways. The first is visual inspection of the angiogram before and after the procedure to determine whether the magnitude of the stenosis (percent stenosis relative to the normal vessel diameter) has been reduced. The degree of stenosis of each lesion can be quantified with calipers, electronic quantitative analysis or graded semi-quantitatively using the following scale: grade 1, no (<10%) stenosis; grade 2, mild (10%–49%) stenosis; grade 3, moderate (50% to 69%) stenosis; and grade 4, severe (70%–99%) stenosis.^{76,77} A second approach is to measure the intragraft pressure before and after the procedure and normalize it for the systemic blood pressure. A third approach is to measure the change in access blood flow before and after the procedure. Each of these measures has been shown to be predictive of the primary patency of AV grafts following elective angioplasty. In one large series, following elective angioplasty, the median intervention-free survival of AV graft with no residual stenosis was 6.9 months, as compared with 4.6 months with any degree of residual stenosis.⁷⁷ Similarly, the median primary patency of AV grafts following angioplasty was inversely related to the intragraft-to-systemic pressure ratio, being 7.6, 6.9, and 5.6 months, when this ratio was <0.4, 0.4 to 0.6, and >0.6, respectively.⁷⁷ Finally, a failure to significantly increase the access blood flow after angioplasty is observed in 20% of AV grafts at 1 week and in 40% by 1 month,^{85,86} confirming the short-lived benefit of this intervention.

Technical procedure: percutaneous graft angioplasty

A digitally subtracted antegrade angiogram is performed to visualize the entire access circuit. The presence or absence of stenotic lesions and their number and location are assessed, and arterial anastomosis, intragraft, venous anastomosis, and draining vein and central vein lesions

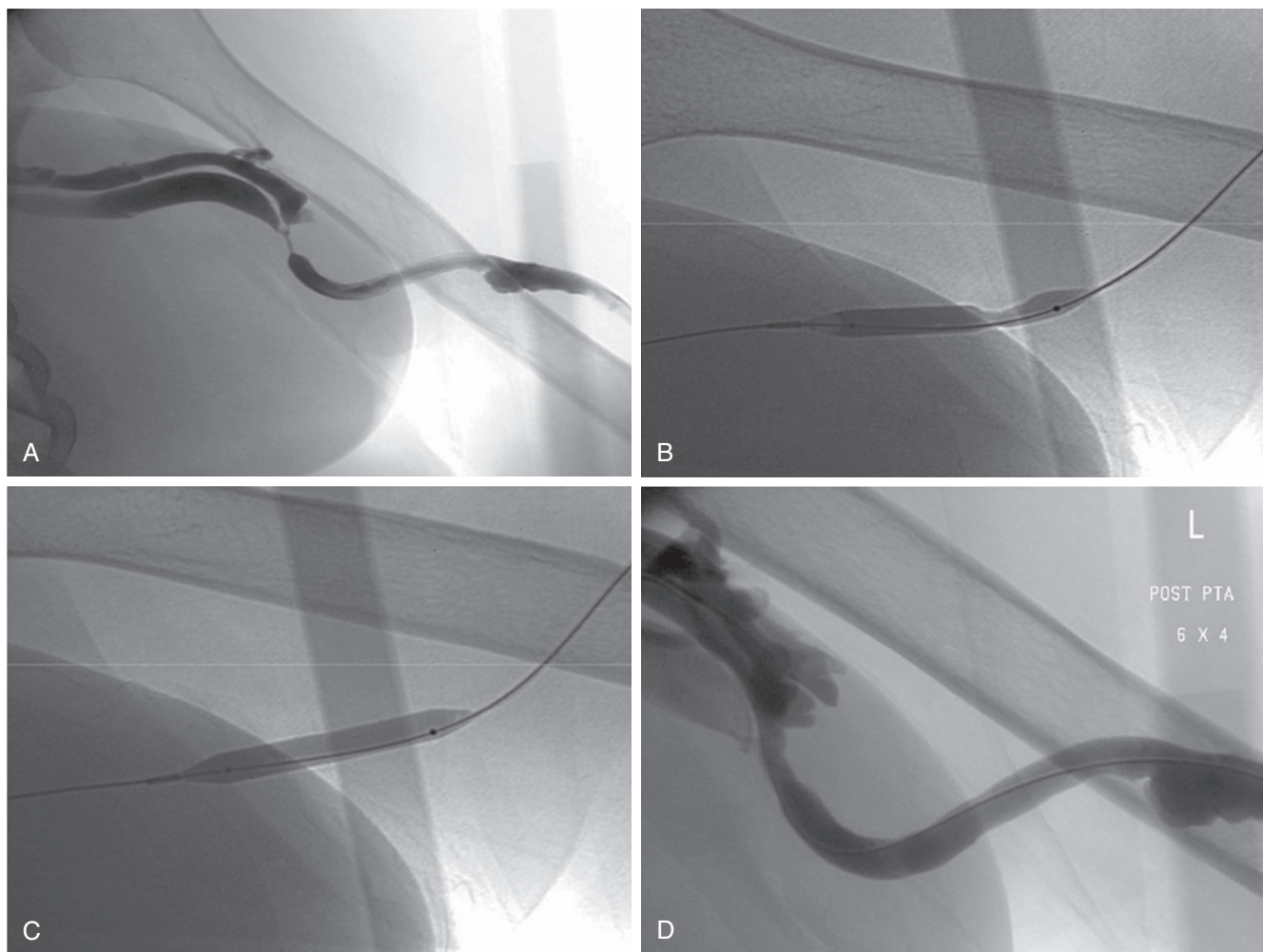


Fig. 67.6 (A) Left upper arm arteriovenous (AV) graft angiogram showing a severe (95%) stenotic lesion at the level of the venous anastomosis. (B) Left upper arm AV graft stenotic lesion at the venous anastomosis with the angioplasty balloon partially inflated. (C) Left upper arm AV graft stenotic lesion at the venous anastomosis with the angioplasty balloon fully inflated. (D) Final postangioplasty left upper arm AV graft angiogram showing a treated lesion with minimal residual stenosis.

are documented. The degree of stenosis of each lesion is quantified with calipers or graded semiquantitatively.^{76,77} Lesions with at least 50% stenosis are considered to be hemodynamically significant and undergo angioplasty using an appropriately sized balloon (Fig. 67.6A–67.6D and see Fig. 67.7A–67.7C). The majority of anastomotic lesions require higher pressure than those required for peripheral arterial angioplasty. Therefore high pressure balloons with minimal burst pressure >25 atmospheres are routinely used.¹⁰³ If a residual (>30%) stenosis is found, prolonged angioplasty (2- to 3-minute inflation), higher-pressure balloons (up to 30 atmospheres), and occasionally covered stent deployment may be required to treat these lesions. The main complication of this procedure is vessel extravasation and/or rupture of the vessel during angioplasty. Most of these extravasations are controlled with balloon inflation for prolonged time and occasionally require covered stent deployment. Surgical repair is rarely indicated and usually reserved for ruptures that are not controlled with stent placement and associated with major local hematoma.

PATHOGENESIS AND TREATMENT OF ARTERIOVENOUS GRAFT DYSFUNCTION

The pathophysiology of AV graft stenosis involves proliferation of vascular smooth muscle cells (neointimal hyperplasia), with progressive encroachment of the lesion into the AV graft lumen.⁷³ There has been ongoing interest in pharmacologic approaches to prevention of neointimal hyperplasia.¹⁰⁴ Two small, single-center, randomized clinical trials have documented a beneficial effect of dipyridamole and fish oil in preventing AV graft thrombosis.^{105,106} A multicenter, randomized, double-blind study compared clopidogrel plus aspirin with placebo for prevention of AV graft thrombosis. The study was terminated early due to an excess of bleeding complications in the intervention group; there was no difference in the rate of AV graft thrombosis between the two randomized groups.¹⁰⁷ Similarly, a randomized clinical trial found that low-intensity warfarin posed a substantial risk of major hemorrhagic complications without reducing the frequency of AV graft thrombosis.¹⁰⁸

The Dialysis Access Consortium was a larger multicenter, randomized, double-blinded clinical trial comparing

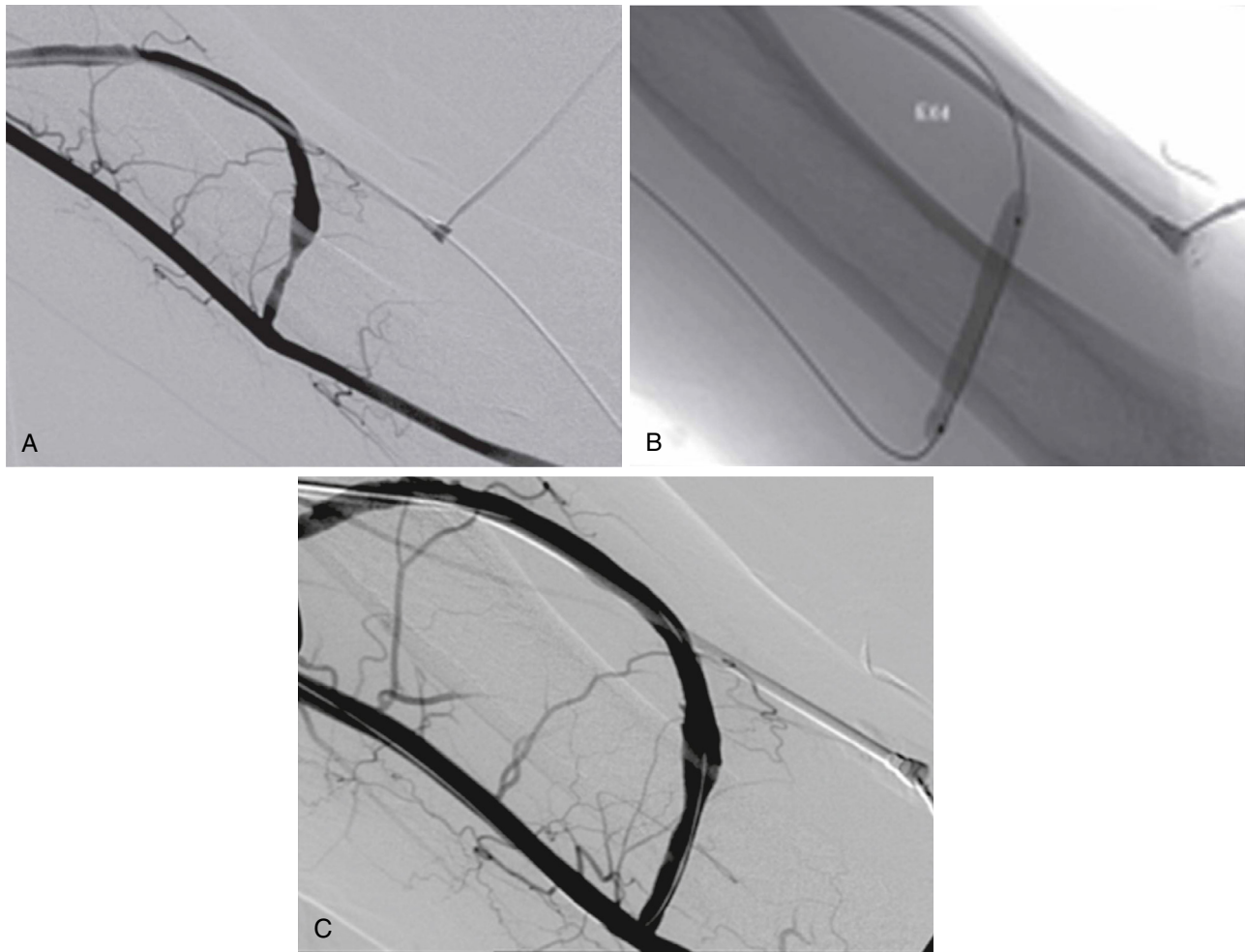


Fig. 67.7 (A) Digital subtraction angiography (DSA) of a left upper arm AV graft showing a moderate stenotic lesion at the level of the arterial limb of the graft. (B) Spot film showing the stenotic lesion being angioplastied. (C) Final postangioplasty DSA showing excellent results.

long-acting dipyridamole + low-dose aspirin (Aggrenox) with placebo in prevention of AV graft failure.¹⁰⁰ There was a modest but statistically significant improvement in primary unassisted AV graft survival in patients receiving dipyridamole-aspirin, as compared with those treated with placebo (28% vs. 23%, at 1 year $P = 0.03$). Cumulative AV graft survival did not differ between the two treatment arms.^{24,100} Another double-blind, randomized, controlled study evaluating fish oil on PTFE AV graft patency and cardiovascular outcomes reported that fish oil did not decrease loss of native patency (time from creation to first intervention) of AV grafts at 12 months but did show improvement in overall AV graft patency and cardiovascular outcomes.¹⁰⁹ In summary, based on available evidence, the use of combination of aspirin and dipyridamole to improve the graft primary patency should be individualized on the basis of patient risk-benefit analysis. Additionally, the use of oral fish oil may reduce the incidence of thrombosis in patients with newly created AV grafts.^{24,104} Finally, to improve the patency of AV grafts following angioplasty, stent deployment was investigated. It was postulated that the rigid scaffold of the stent could potentially keep the vascular lumen open. The use of stents in dialysis access is discussed later.¹¹⁰

ARTERIOVENOUS FISTULA MALFUNCTION

SALVAGE OF IMMATURE AV FISTULAS

Compared with AV grafts, AV fistulas require a much lower frequency of intervention (angioplasty or thrombectomy) to maintain their long-term patency for dialysis.²⁰ However, AV fistulas have a substantially higher primary failure rate (AV fistulas that are never usable for dialysis). Similar to AV grafts, the rate of angioplasty utilization increased from 0.16 to 0.47 (rate per patient year) in U.S. Renal Data System data,²³ likely due to the increasing frequency of treating primary AV fistula failures. The proportion of AV fistulas with primary failure has ranged from 20% to 60% in multiple series, even when routine preoperative vascular mapping has been employed.^{20,65,111} Primary AV fistula failures fall into two major categories: early thrombosis and failure to mature.¹¹²⁻¹¹⁴ Early thrombosis refers to AV fistulas that thrombose within 3 months of the creation, before they have been used for dialysis. Failure to mature refers to AV fistulas that never develop adequately to be cannulated reproducibly for dialysis. Nonmaturation is less common with upper arm than forearm AV fistulas.²⁰ Among upper arm AV fistulas, nonmaturation is less likely with transposed brachio basilic than brachiocephalic AV fistulas.¹¹⁵

Table 67.9 Effect of Salvage Procedures on Immature Fistulas

Reference	No. of Patients	Type of Intervention	Percent Usable for Dialysis	Primary Patency at 1 Year
Beathard, 1999 ¹¹⁸	63	PTA, Vein ligation	82	75%
Turmel-Rodrigues, 2001 ¹²⁰	69	PTA, Vein ligation	97	39%
Miller, 2003 ¹¹⁴	41	PTA Vein ligation	44	N/A
Beathard, 2003 ¹¹⁹	100	Surgical revision PTA, Vein ligation	92	68%
Asif, 2005 ¹¹⁷	24	PTA, Vein ligation	92	N/A
Nassar, 2006 ¹²¹	119	PTA, Vein ligation	83	65%
Singh, 2008 ¹²³	32	PTA Vein ligation	78	NA
Han, 2013 ¹⁸⁵	141	Surgical revision PTA	87	72%

NA, Not available; PTA, percutaneous transluminal angioplasty.

In some patients, maturation of the AV fistula can be assessed easily by clinical evaluation by the nephrologist, surgeon, or an experienced dialysis nurse. In less straightforward cases, duplex ultrasound may be useful in predicting whether a new AV fistula can be used successfully for hemodialysis. One pilot study used a combination of two simple sonographic criteria to assess AV fistula maturation: AV fistula diameter and access blood flow.²⁷ When the ultrasound documented a draining vein diameter ≥ 4 mm and an access blood flow ≥ 500 mL/min, 95% of the AV fistulas were subsequently usable for dialysis. In contrast, when no criterion was met, only 33% of the AV fistulas achieved adequacy for dialysis. The likelihood of AV fistula adequacy for dialysis was intermediate (~70%) when only one of the two criteria was met.

Failure of an AV fistula to mature can be related to one of several anatomic defects, which can be identified either by sonography or angiography.¹¹⁶ The main causes of fistula failure to mature are stenosis at the arterial anastomosis/juxta-anastomotic segment or in the draining vein and the presence of large accessory veins that affect fistula blood flow. In most cases, these anatomic problems can be corrected percutaneously or surgically. Stenotic lesions can be treated by angioplasty or surgical revision. Superficial side branches can be ligated by a suture through the skin; deeper branches can be embolized.

It is worth mentioning that in obese patients, the AV fistula may have adequate caliber and blood flow but is too deep to be cannulated. This may require surgical superficialization to facilitate AV fistula cannulation.

In immature AV fistulas with one or more of the anatomic lesions described, specific interventions to correct the underlying lesion may promote subsequent AV fistula maturation. Several published series have evaluated the ability to salvage immature AV fistulas such that they are subsequently usable for dialysis. A number of studies using only radiographic procedures (angioplasty of stenotic

lesions or obliteration of side branches) in immature AV fistulas have shown a high success rate (Table 67.9).¹¹⁷⁻¹²² An initial salvage (ability to use the AV fistula for dialysis) was accomplished in 80% to 90% of patients, with a subsequent 1-year primary patency of 39% to 75%. A study using a combination of radiologic and surgical salvage procedures in an unselected dialysis population reported a more modest salvage rate of 44%.¹¹⁴ The frequency of salvage procedures for immature AV fistulas in this study was twice as high in women than men. Retrospective analysis of immature AV fistulas with an anatomic abnormality that were corrected with angioplasty or surgically found that these were more likely to be usable dialysis, compared with similar fistulas that did not undergo a salvage procedure.¹²³ Balloon-assisted maturation may enhance the maturity rate in these AV fistulas.¹²⁴ The premise of balloon-assisted maturation is to perform repeated long-segmental sequential angioplasty of the whole access to promote maturity.¹²⁴ Further, balloon-assisted maturation was used intraoperatively during fistula creation in patients with small arteries and veins (artery, <2 mm; vein, <2.5 mm).^{125,126} Overall, these techniques were not compared with a control group and fistulas generally required further angioplasty after maturity to maintain patency. Importantly, immature AV fistulas requiring interventions to achieve maturity had significantly shorter cumulative patency and required more interventions to maintain patency, compared with AV fistulas achieving maturation without interventions^{127,128} (Fig. 67.8). Randomized studies comparing different types of interventions or different timing of interventions in terms of their effect on AV fistula maturation are generally lacking. In practice, proactive percutaneous or surgical interventions are attempted to salvage these immature accesses.

There has also been interest in pharmacologic interventions to promote AV fistula maturation. A randomized study of 877 U.S. patients undergoing creation

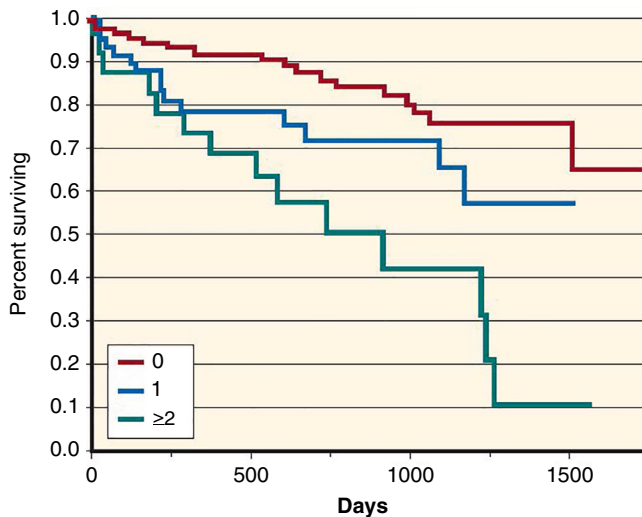


Fig. 67.8 Interventions to promote arteriovenous fistula maturation. Cumulative survival, defined from the time of access cannulation to permanent failure, is shorter in patients receiving two or more interventions before arteriovenous fistula maturation compared with those with zero interventions (hazard ratio, 2.07; 95% confidence interval, 1.21 to 2.94; $P = 0.0001$). (Reproduced from Lee T, Ullah A, Allon M et al. Decreased cumulative access survival in arteriovenous fistulas requiring interventions to promote maturation. *Clin J Am Soc Nephrol*. 6:575–581, 2011 with permission from the American Society of Nephrology.)

of a new AV fistula allocated patients to receive clopidogrel or placebo for the initial 6 weeks.⁵⁰ While clopidogrel significantly decreased the frequency of AV fistula thrombosis within 6 weeks, there was no effect on access maturation.⁵⁰ Subsequently, another randomized study from Australia and New Zealand of 567 patients reported that neither fish oil nor low-dose aspirin prevented new AV fistula failure.¹²⁹ There is some evidence to support the role of far-infrared therapy in improving the primary patency of AVF. The use of this modality was evaluated in four studies in Taiwan ($N = 763$) and reported higher primary patency rate at 1 year of follow-up.²⁴ The use of fish oil or aspirin did not show to prevent AV fistula flow dysfunction. There is inadequate evidence to support the use of simvastatin, ezetimibe, or clopidogrel-prostacyclin to reduce fistula thrombosis or improve primary failure.²⁴

Technical aspects of salvaging immature arteriovenous fistulas

Percutaneous transluminal angioplasty of arteriovenous fistulas. The stenosis at the juxta anastomotic segment and within the venous outflow can be treated with sequential balloon dilatations over one or more angioplasty sessions. Generally, the fistula is cannulated in a retrograde manner using ultrasonography guidance and the feeding artery is cannulated. Subtracted digital angiograms of the dialysis circuit usually delineates the culprit lesion(s) that are treated with appropriately sized balloons (Fig. 67.9A and 67.9B). In practice, these patients are brought back in 2 to 4 weeks for physical examination and repeat angiogram as clinically warranted. Once the fistula is deemed suitable for cannulation, it is highly recommended that an experienced nurse cannulates the new access using one short needle and then advance to two needles as feasible.

Ligation of accessory veins. Accessory veins that affect the fistula maturity can be treated by either surgical ligation or endovascular coil deployment. Accessory veins are treated depending on size, location, and number (Fig. 67.10). While superficial and deep large accessory veins are usually ligated surgically, vessel embolization using coil deployment is used to treat small deep accessory veins¹³⁰ (Fig. 67.11). To accomplish this, the fistula is accessed and a selective cannulation and an angiogram are performed on the intended veins, followed by deploying appropriately sized coils. A final angiogram of the AV fistula is performed to ascertain proper coil deployment and occlusion of all collateral veins.

Although the frequency of interventions is several-fold lower in AV fistulas than grafts,²⁰ AVFs are also susceptible to developing stenosis and thrombosis. Most studies have documented a comparable primary patency of fistulas and grafts following elective PTA (Table 67.10),^{76,80,90,102} although one study observed a higher primary patency in fistulas.¹³¹ As is the case with angioplasty of grafts, the primary patency of fistulas following PTA is inversely related to the magnitude of postangioplasty stenosis, as well as the magnitude of the postangioplasty intraaccess to systemic pressure ratio.⁷⁶

After performing the initial digitally subtracted angiogram of the access circuit, PTA is done on the hemodynamically significant lesions ($>50\%$ stenosis). A final digitally subtracted angiogram is performed to assess for residual stenosis and requirement for further treatment of the stenotic lesion (Fig. 67.12). In some centers, intrafistula pressure and systemic pressure are measured before and immediately after the intervention to assess the efficacy of angioplasty; a reduction in the intrafistula to systemic pressure ratio is used to confirm hemodynamic improvement. The major complications of this procedure are vessel extravasations and rupture of the vessel after the angioplasty. As in AV grafts, most of these complications are controlled with prolonged balloon tamponade at the extravasation site. Stent deployment is indicated occasionally, and surgical referral is rarely needed (Fig. 67.13).

ANEURYSMAL FORMATIONS OF THE ARTERIOVENOUS ACCESS

Highly matured and developed AVFs dilate and elongate significantly over the years, forming aneurysmal dilatation. With fixed points at the wrist and elbow or elbow and shoulder, AVFs become tortuous and form significant kinks at certain points between dilated segments at certain intervals, which may appear as significant stenoses. Many of these are not flow limiting, while some are truly flow limiting. These are unlikely to respond to PTA and may require surgical intervention. Clinical factors and major comorbidities such as age, race, diabetes, presence of peripheral vascular disease, access location, and number of stenotic sites do not predict the likelihood of vascular access patency after PTA.⁷⁶

Pseudoaneurysmal formations in AVGs are related to repeated cannulation of the graft material, leading to compromise of the wall integrity. The presence of outflow vein stenoses and/or excessive access flow may contribute to the development of aneurysms and pseudoaneurysms and their related complications.^{132,133} Conservative clinical observation is recommended in the following scenarios:

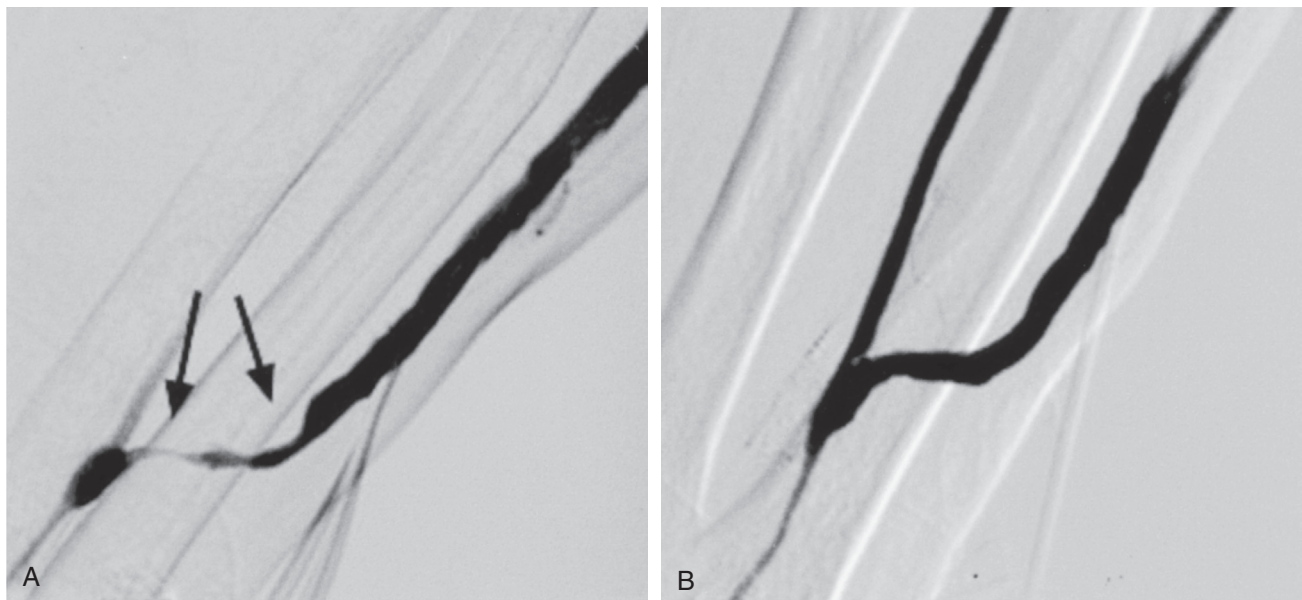


Fig. 67.9 (A) Fistula salvage. Digital subtraction angiography (DSA) of a radiocephalic arteriovenous fistula showing a severe stenotic lesion at the level of the juxta arterial anastomosis. (B) Fistula salvage: Postangioplasty DSA of a radiocephalic fistula showing radiologic improvement of the stenotic lesion at the level of the juxta arterial anastomosis.

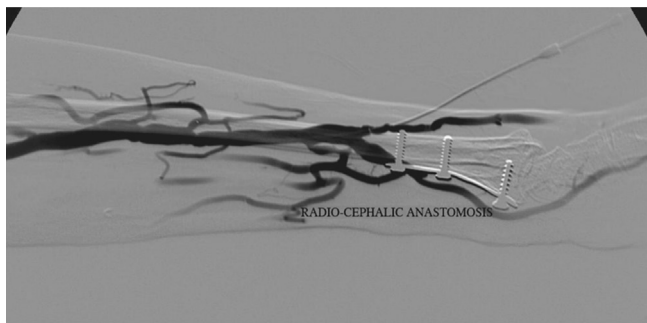


Fig. 67.10 Digital subtraction angiography of a left radiocephalic arteriovenous fistula showing multiple collaterals. There is a metallic plate from a prior open reduction and internal fixation of a radius bone fracture.

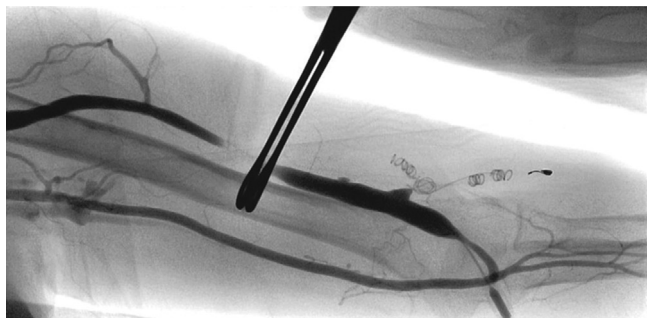


Fig. 67.11 Coil deployed in a collateral vein of an upper arm arteriovenous fistula.

lack of clinical manifestations (such as excessive bleeding after dialysis); the skin integrity of these formations is intact; and minimal risk of rupture.¹³³ Some experts recommend surgical repair as the first line of therapy for symptomatic aneurysms and pseudoaneurysms. In a series of 209 patients who had 24 AVGs and 185 AVFs, Wang and colleagues¹³²

Table 67.10 Primary Patency After Elective Angioplasty: Fistulas Versus Grafts		
Reference	Primary Access Patency at 6 Months	
	Grafts	Fistulas
Safa, 1996 ⁸⁰	43%	47%
Turmel-Rodrigues, 2000 ¹⁰²	53%	67%
McCarley, 2001 ⁹⁰	37%	34%
Van der Linden, 2002 ¹³¹	25%	50%
Maya, 2004 ⁷⁶	52%	55%

used a staged surgical repair using pneumatic tourniquet and found that the assisted primary patency rate at 1 and 3 years was 96% and 87%, respectively, with only 2% of the cohort required bridging CVC (Fig. 67.14). Stenting is not recommended due to the risk of infection, compromising cannulation segment and stent fractures.

ENDOVASCULAR MANAGEMENT OF THROMBOSED DIALYSIS ACCESS

Substantial morbidity, access failure, and increased health care cost are associated with access thrombosis.⁴⁷ While the majority of AV graft failures are due to thrombosis that is related to an underlying stenosis at the venous anastomosis, fistula thrombosis is usually caused by inflow stenotic lesion or outflow severe stenosis caused by hematomas.^{75,77} As discussed earlier, several mechanisms including neointimal hyperplasia are involved in the pathogenesis of dialysis access thrombosis that lead to luminal narrowing and subsequent thrombosis (Fig. 67.15). In mature dialysis access, the repeated needle cannulation and balloon angioplasty can

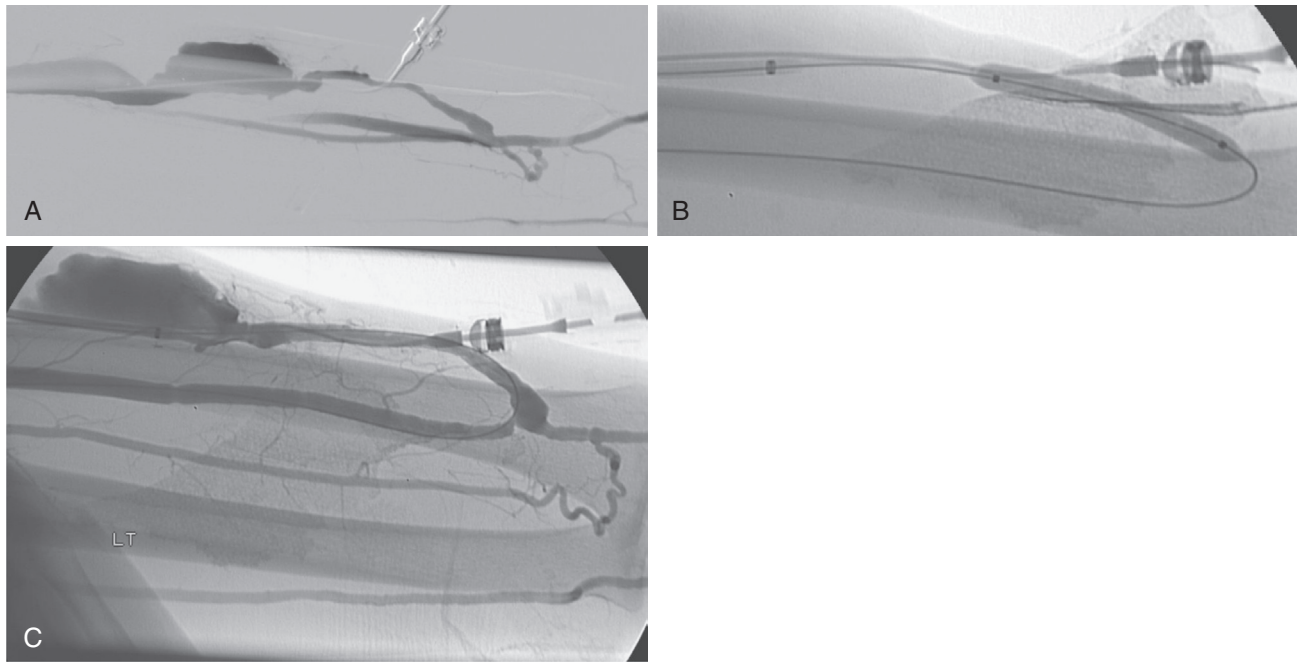


Fig. 67.12 (A) Digital subtraction angiography (DSA) of a left radiocephalic arteriovenous (AV) fistula showing a long severe segment of stenosis distal to the arterial anastomosis followed by a pseudoaneurysm of the fistula. (B) Spot film of a left radiocephalic AV fistula showing the segment of stenosis being angioplastied. (C) Postangioplasty DSA shows a successful treatment, and the pseudoaneurysm is unchanged.

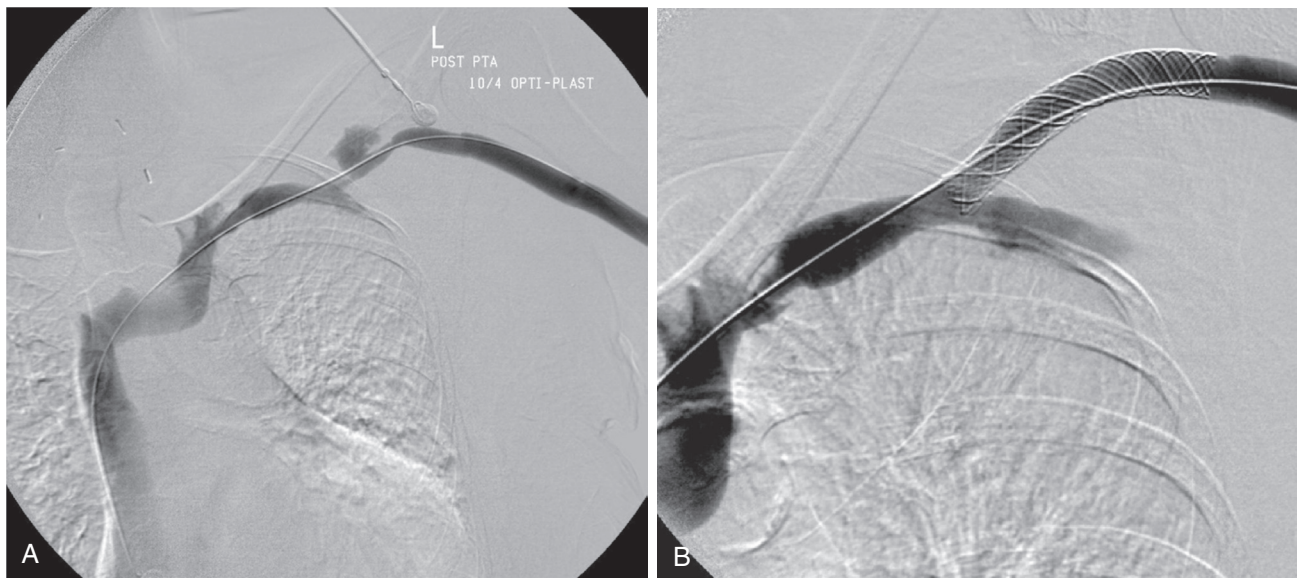


Fig. 67.13 (A) Angioplasty complication. Digital subtraction angiography (DSA) showing a rupture of the left cephalic vein after aggressive percutaneous transluminal angioplasty. There is a coexisting stenosis of the left subclavian. (B) Angioplasty complication: DSA showing salvage and correction of the complication by deploying a covered stent (wall-graft).

propagate further endothelial damage and subsequent luminal stenosis. The most common cause of AV fistula thrombosis is an underlying stenotic lesion in the venous outflow circulation (either peripherally or centrally). Less common causes include needle infiltration,¹³⁴ excessive manual pressure for hemostasis at the needle insertion site, or severe and prolonged hypotension (Fig. 67.16).¹³⁵ Unlike AVFs that can remain patent with lower access flow rate of 300 mL/min, the risk of AVG thrombosis increases when the access blood flow drops <600 mL/min. This differential threshold of thrombosis between fistulas and grafts is mainly

attributed to the lack of endothelial layer in AVGs (Fig. 67.17).^{136,137}

In general, the fundamental principle of access thrombectomy revolves around removing the clot burden from the access circuit and correcting the culprit lesion as detailed in the following paragraphs.⁴⁷

DIAGNOSIS OF DIALYSIS ACCESS THROMBOSIS

The diagnosis of dialysis access thrombosis is usually made clinically; the absence of thrill bruit or pulse during physical examination indicates access thrombosis. Access



Fig. 67.14 Aneurysm repair. (A) Aneurysm formation presented with wall thinning and bleeding. (B) The aneurysm is marked and an incision was made. (C) Using a pneumatic tourniquet, the aneurysmal tissue was removed. (D) Closing the incision with nonabsorbable sutures. (E) Closing the skin incision. (F) Appearance of the fistula after repair. (From Wang S, Wang MS. Successful use of partial aneurysmectomy and repair approach for managing complications of arteriovenous fistulas and grafts. *J Vasc Surg.* 2017;66[2]:545–553.)

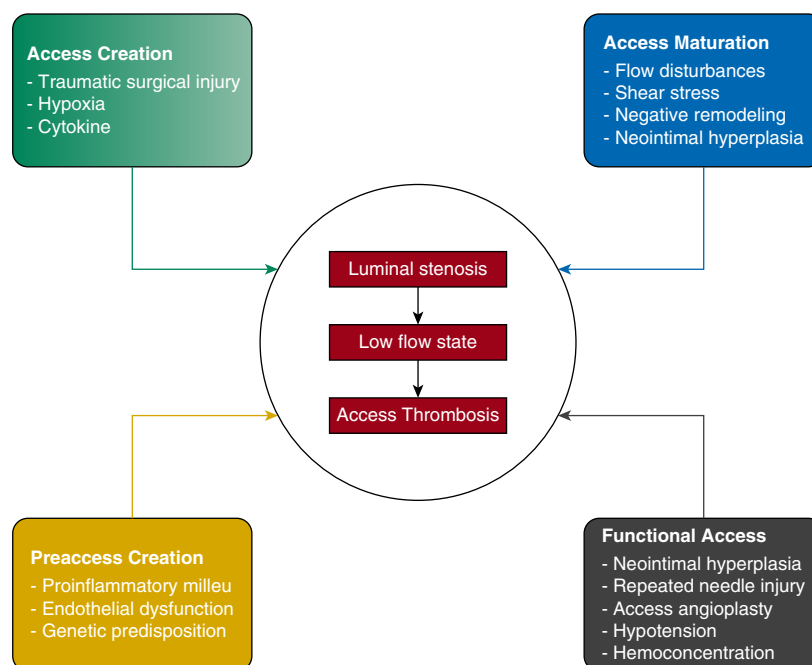


Fig. 67.15 Factors that contribute to pathogenesis of dialysis access thrombosis. (From Almeihmi A, Al-Balas A. The clotted access. In: Illig KA, Scher LA, Ross JR, ed(s). *Principles of Dialysis Access.* Springer; 2024:419–434.)

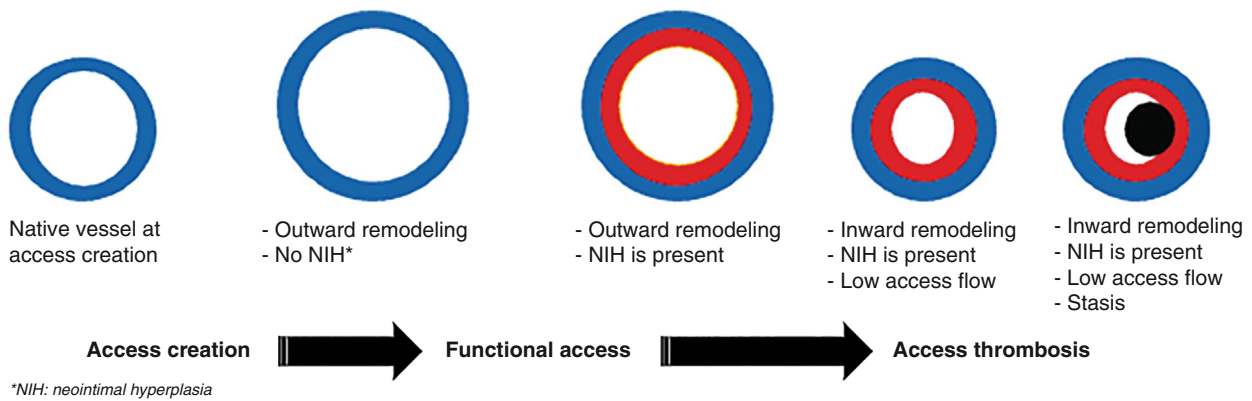
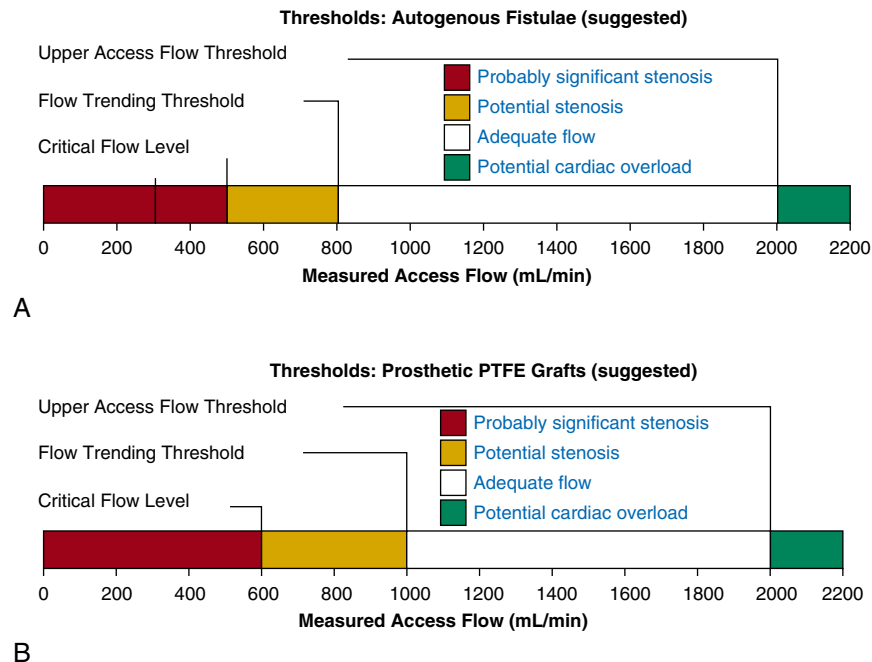


Fig. 67.16 Pathologic stages of dialysis access thrombosis.

Fig. 67.17 (A and B) Relationship between access flow rate and risk of thrombosis. (From Almekhi A, Sheta M, Abaza M, et al. Endovascular management of thrombosed dialysis vascular circuits. *Semin Interv Radiol*. 2022;39:14–22.)

ultrasonography is rarely needed for diagnosis. Once the diagnosis is made, access thrombectomy is recommended as soon as possible to improve access survival and avoid CVC use.⁴⁷ Localized pain and tenderness can be encountered over the thrombosed AVF site due to local inflammatory thrombophlebitis. If these findings are seen in a thrombosed AVG, it usually denotes graft infection.¹³⁸

CONTRAINDICATIONS OF DIALYSIS ACCESS THROMBECTOMY

There are few contraindications to perform endovascular thrombectomy in a timely fashion. These contraindications are classified into absolute, relative, and temporary. Access infection, pulmonary hypertension, recent access creation, severe ipsilateral steal syndrome, and right-to-left cardiac shunts are considered absolute contraindications.¹³⁶ Performing endovascular thrombectomy in a recently created vascular access carries a risk of rupture of the fresh surgical anastomotic site. As such, these procedures should

be done by experienced interventionalists or vascular surgeons.⁴⁷ Fluid overload, hyperkalemia ($K > 6 \text{ mEq/L}$), and hemodynamic instability are regarded as temporary contraindications. In these cases, a temporary dialysis catheter will be required to perform dialysis therapy before access thrombectomy. Recent intracranial hemorrhage, mega fistula, contrast allergy, and cardiac arrest are considered relative contraindications and should be left to the operator discretion and consideration of the risk of the required anticoagulation.¹³⁶

DIALYSIS ACCESS THROMBECTOMY PROCEDURE

Currently, endovascular thrombectomy is considered the procedure of choice to treat dialysis access thrombosis. The time between vascular access thrombosis and reestablishment of access flow is an independent predictor of vascular access survival; therefore it is imperative to perform thrombectomy as soon as possible.⁴⁷ When performing access thrombectomy, four fundamental

principles require special attention from the operator. Firstly, the location of the thrombus and the clot burden can vary between fistulas and grafts. Whereas the thrombus generally occupies the entire length of the synthetic graft, in the thrombosed AVF, the thrombus is usually localized at the juxta-anastomotic segment area.¹³⁹ Secondly, thrombectomy of AVFs is more challenging compared with AVGs. The reason for this difference is that AVF thrombosis can trigger an inflammatory response that leads to the adherence of the thrombus to the vessel wall. Consequently, the success rate for AVF thrombectomy declines significantly if thrombectomy is delayed by more than 2 to 3 days. However, AVG thrombectomy can be performed successfully up to 2 weeks following its thrombosis.¹³⁹ Thirdly, thrombosis of the dialysis access within the first month of creation requires surgical thrombectomy due to the inherent risk of rupturing the suture line with balloon inflation. Finally, recurrent AVGs thrombosis (>3 episodes of thrombosis within 3 months) and AVF thrombosis soon after its creation often require surgical revision.⁴⁷

OUTCOMES OF DIALYSIS ACCESS THROMBECTOMY

The primary patency of AV grafts after thrombolysis or thrombectomy and angioplasty (see Table 67.9) ranges from 30% to 63% at 3 months and 11% to 39% at 6 months.^{77,102,140-147} These outcomes are considerably worse than the primary patency observed after elective angioplasty (Table 67.11), which is 71% to 85% at 3 months and 47% to 63% at 6 months.^{76-78,80,101,102}

The primary AV graft patency is similar for mechanical thrombectomy and pharmacomechanical thrombectomy.¹⁴² A large series comparing the outcomes of both types of radiologic procedures at one institution found that the primary patency was only 30% at 3 months for thrombosed AV grafts, as compared with 71% for patent AV grafts undergoing elective angioplasty (Fig. 67.18A).⁷⁷ The discrepancy was still apparent when the analysis was restricted to the subset of procedures in which there was no residual stenosis, with a median primary patency of 2.5 months after thrombectomy, as compared with 6.9 months after elective angioplasty (Fig. 67.18B).⁷⁷

The duration of AV graft patency following thrombectomy does not appear to differ among patients with and without diabetes mellitus. Arteriovenous graft patency is also generally unrelated by the AV graft location or the number of concurrent AV graft stenoses found.⁷⁷ However, similar to the observations obtained after elective angioplasty, the primary patency of AV grafts after thrombectomy is inversely proportional to the magnitude of residual stenosis at the end of the procedure.⁷⁷

Salvaging thrombosed AV fistulas is one of the most challenging aspects in interventional nephrology.¹⁴⁸ Thrombectomy of aneurysmally dilated AV fistulas is the most challenging technically. Several series have reported on the outcomes of radiologic thrombectomy of AV fistulas.^{102,148,149-155} The immediate technical success has

Table 67.11 Primary Graft Patency After Thrombectomy

Reference	No. of Procedures	Primary Patency at:	
		3 Months	6 Months
Valji, 1991 ¹⁴⁰	121	53	34
Trerotola, 1994 ¹⁴¹	34	45	19
Beathard, 1994 ¹⁴²	55 mech	37	
	48 pharm	46	
Cohen, 1994 ¹⁴³	135	33	25
Sands, 1994 ¹⁴⁴	71		11
Beathard, 1995 ¹⁴⁵	425	50	33
Beathard, 1996 ¹⁴⁶	1176	52	39
Trerotola, 1998 ¹⁴⁷	112	40	25
Turmel-Rodrigues, 2000 ¹⁰²	58	63	32
Lilly, 2001 ⁷⁷	326	30	19

been fairly high, ranging from 73% to 93%. The primary patency of these AV fistulas following thrombectomy has ranged from 27% to 81% at 6 months, and 18% to 70% at 1 year (Table 67.12).^{102,149,152-156} In one study, the primary patency following thrombectomy was lower in upper arm AV fistulas, as compared with those in the forearm.¹⁰² However, with additional interventions, the secondary patency of these AV fistulas ranges from 44% to 93% at 1 year. A study of 140 consecutive patients, over a 2-year period, with thrombosed immature AV fistulas who underwent salvage thrombectomy procedures found that thrombectomy was successful in 119 (85%) immature clotted AV fistulas, and hemodialysis adequacy was achieved in 111 (79%).¹⁵⁶ The average maturation time from thrombectomy to cannulation for dialysis was 46.4 days, with an average of 2.64 interventions per patient. There were five (3.5%) cases of angioplasty-induced rupture, all of which were treated with stent placement. Clinically significant pseudoaneurysm formation occurred in four (2.8%) patients. At 12 months, secondary access patency of the salvaged accesses was 90%. Considering that the alternative would be to abandon the thrombosed fistula and proceed with placement of a new AV fistula, concerted efforts aiming to salvage thrombosed AV fistulas are extremely worthwhile.

Technical Procedure: Percutaneous Dialysis Access (AVG And AVF) Thrombectomy

Thrombectomy of dialysis access is a multistep procedure that aims to restore the blood flow within the access circuit by utilizing a combination of thrombolysis and mechanical thrombectomy using different devices, balloons, and stents as clinically indicated.

The fundamental principles of access thrombectomy center around 1. removal of the thrombus from the access conduit, and 2. correcting the underlying culprit stenosis of the access⁴⁷ (Fig. 67.19).

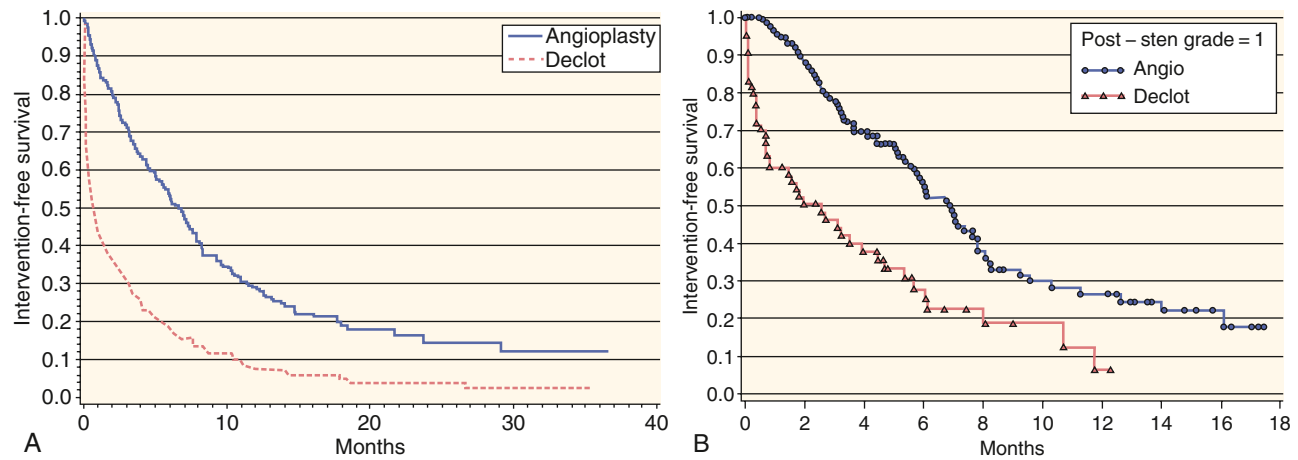


Fig. 67.18 (A) Intervention-free graft survival following elective angioplasty (solid line) or thrombectomy + angioplasty (dotted line). Graft survival was calculated from the date of the initial intervention to the date of the next intervention (angioplasty, declot, or surgical revision). $P < 0.001$ for the comparison between the two groups. (B) Intervention-free graft survival following elective angioplasty (circles) or thrombectomy + angioplasty (triangles) in the subset of procedures with no residual stenosis. Graft survival was calculated from the date of the initial intervention to the date of the next intervention (angioplasty, declot, or surgical revision). $P < 0.001$ for the comparison between the two groups. (A and B, Reproduced from Lilly RZ, Carlton D, Barker J, et al. Predictors of arteriovenous graft patency after radiologic intervention in hemodialysis patients. *Am J Kidney Dis.* 2001;37(5):945–953.)

Table 67.12 Primary Fistula Patency After Thrombectomy

Reference	No. of Procedures	Primary Patency at:	
		6 Months	12 Months
Haage, 2000 ¹⁴⁹	54		27
Turmel-Rodrigues, 2000 ¹⁰²	54 FA 9 UA	74 27	47 27
Rajan, 2002 ¹⁵³	30	28	24
Liang, 2002 ¹⁵²	42	81	70
Shatsky, 2005 ¹⁵⁴	44	38	18
Jain, 2008 ¹⁵⁵	41	20	—
Miller, 2011 ¹⁵⁶	140	—	59

Series with fewer than 25 procedures not included.

Removal of the Thrombus from the Access Conduit

After obtaining informed consent, an intravenous line should be established to administer agents such as midazolam and/or fentanyl for conscious sedation. The endovascular thrombectomy procedure is performed in the interventional suite, where the patient's blood pressure, heart rate, oxygen saturation, and cardiac rhythm are continuously monitored. Local anesthesia is achieved with 1% to 2% lidocaine and 2000 to 5000 IU of heparin is given systematically.⁴⁷ Two antegrade and retrograde access sheaths (6 or 7 Fr) are inserted facing each other without overlapping. Occasionally, ultrasound guidance is used to access the AVF/AVG. Next, a pullback angiogram is performed to assess the outflow veins and the level of stenotic lesions.⁴⁷ Most interventionalists currently use a combination of thrombolysis and mechanical thrombectomy to remove the clot burden^{47,157,158} (Fig. 67.20).

Before using thrombolytic agents, it is crucial to assess the patient for the presence of any contraindications to thrombolytic therapy, such as recent surgery, bleeding disorder, recent episode of bleeding, and the presence of severe hypertension.

Several thrombolytic agents are available including urokinase and streptokinase, as well as tissue plasminogen activator (tPA), which is the most used agent in the United States. The technical success of thrombolysis in reestablishing the blood flow in a thrombosed dialysis circuit is modest ranging between 33% and 80%.¹⁵⁷ For this reason, mechanical thrombectomy is often combined with pharmacologic thrombolysis to improve the access patency rate.

Mechanical thrombectomy uses different devices in combination with thrombolytics to reestablish blood flow. In essence, devices of mechanical thrombectomy are divided into two categories: direct wall contact devices and hydrodynamic devices (see Fig. 67.19).

1. **Direct Wall Contact Devices:** Using a balloon, such as Fogarty balloons (Edwards Laboratories, Santa Ana, CA) or a rotational device, such as Arrow-Tretrout (Teleflex, Wayne, PA), and Cleaner XT Rotational Thrombectomy System (Argon Medical Devices, Plano, TX), the luminal clot burden is removed by pullback approach.⁴⁷ The rotational device works by fragmenting the clot within the access circuit via generating a hydrodynamic vortex created by the high speed (800–5000rpm) rotating impella.^{136,159} These devices are inexpensive and disposable. Some of them can be used over the wire. Almekhi and colleagues¹⁵⁸ described the use of Fogarty in combination with a no-wait lysis approach in performing AVG thrombectomy. In this study, the Fogarty balloon was used to remove the arterial plug and establish the blood flow from the feeding artery. Once the arterial flow was established, tPA and heparin were infused and the clot burden was macerated using noncompliant angioplasty balloon. This approach was associated with a high technical success rate of thrombectomy and shorter radiation and procedure time.¹⁵⁸

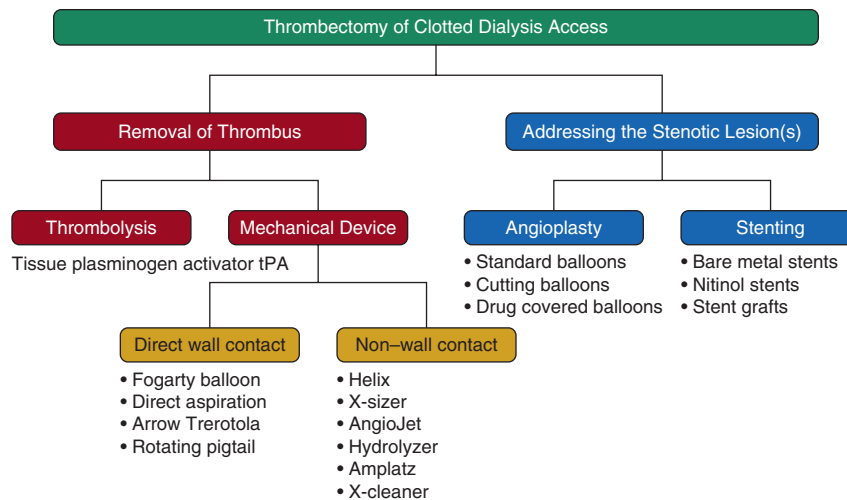


Fig. 67.19 Principles of performing dialysis access thrombectomy. (From Almehehi A, Al-Balas A. The clotted access. In: Illig KA, Scher LA, Ross JR, eds. *Principles of Dialysis Access*. Springer; 2024:419–434.)

2. Hydrodynamic (rheolytic, non-wall contact) Devices: These devices use high-speed rotation to generate a hydrodynamic vortex that disrupts the clot and creates a negative pressure gradient. Subsequently, this negative gradient leads to the removal of the clot microfragments into a collection bag without vessel wall contact. The hydrodynamic vortex results from the high-speed fluid jets that produce the Venturi (rheolytic) effect.^{136,157} Several devices are used, such as AngioJet (Boston Scientific, Marlborough, MA), Oasis catheter (Boston Scientific, Natick, MA), Hydrolyzer (Cordis, Miami, FL), Amplatz thrombectomy device (Microvena, White Bear Lake, MN), and Straub Rotarex catheter (Straub Medical AG, Wangs, Switzerland).⁴⁷ For instance, an AngioJet catheter is a wire-guided catheter that comes in different sizes, 4- to 6-Fr systems. After passing the catheter over a wire beyond the distal end of the thrombus, the catheter is activated and drawn back into the clot. At this point, the catheter generates high-speed saline jets in a retrograde direction that produces a Venturi suction gradient that breaks down the clot burden. Finally, these generated fragments are evacuated into the device. Overall, the success rate of the above devices in reestablishing blood flow in thrombosed dialysis access is more than 90%.^{157,160-162}

Addressing the Underlying Culprit Lesion

Dialysis access thrombosis is primarily caused by an underlying stenosis of access conduit that necessitates angioplasty. Several types of angioplasty balloons have been used in access thrombectomy. These balloons are noncompliant and usually classified as high-pressure, ultra-high pressure, cutting, or drug-coated balloons (DCBs).¹⁶³ Standard high-pressure balloons are commonly used in these procedures and are considered the modality of choice (see Fig. 67.20), although they are associated with poor patency rates and recurrent stenosis. To minimize the effects of balloon angioplasty on endothelial injury, several studies investigated the role of DCBs in managing AVG stenotic lesions: angioplasty with paclitaxel versus standard balloon angioplasty to evaluate the effects of minimizing

endothelial injury following angioplasty.^{164,165} In a multicenter, randomized controlled trial in 285 patients, the use of a Lutonix drug-coated balloon in dysfunctional upper extremity AV fistulas showed no major benefit. The target lesion primary patency at 6 months was 71% for the drug-coated balloon and 63% for the control group ($P = 0.06$).¹⁶⁶ In a meta-analysis that evaluated 14 randomized controlled trials in 1535 patients, which compared DCBs to conventional standard balloons in treating dysfunctional dialysis access, there was no significant difference of the target lesion primary patency between the two groups.¹⁶⁷ The increased cost associated with DCBs may also limit their routine use in clinical practice.^{168,169} Some evidence suggests that stent use in managing dialysis access dysfunction may improve the access patency compared with balloon angioplasty alone (discussed later).¹⁷⁰ However, in-stent stenosis may develop, requiring further angioplasty and on rare occasions stent deployment within the existing stents.⁴⁷

COMPLICATIONS OF DIALYSIS ACCESS THROMBECTOMY

Complications encountered during thrombectomy may include vessel rupture, arterial embolism, pulmonary embolism, and periaclous hematoma. The incidence of arterial embolism after thrombectomy is low, ranging between 0.4% and 7%.¹⁵⁷ This complication can be managed with balloon embolectomy, selective thrombolytic infusion with tPA, back bleeding, and surgical embolectomy. Precautions to limit the risk of arterial embolism include minimizing manual compression of the vascular access, avoiding flushing of the sheaths after placement, avoiding reflux angiograms, and completely deflating the balloon when crossing the arterial anastomosis into the feeding artery.^{47,136} Hematoma formation after thrombectomy is encountered in <3% of cases and is usually treated conservatively. Large hematomas may require external pressure, stent placement, and in some cases surgical evacuation.¹⁷¹ The incidence of clinically significant pulmonary embolism during or after thrombectomy is low.⁴⁷

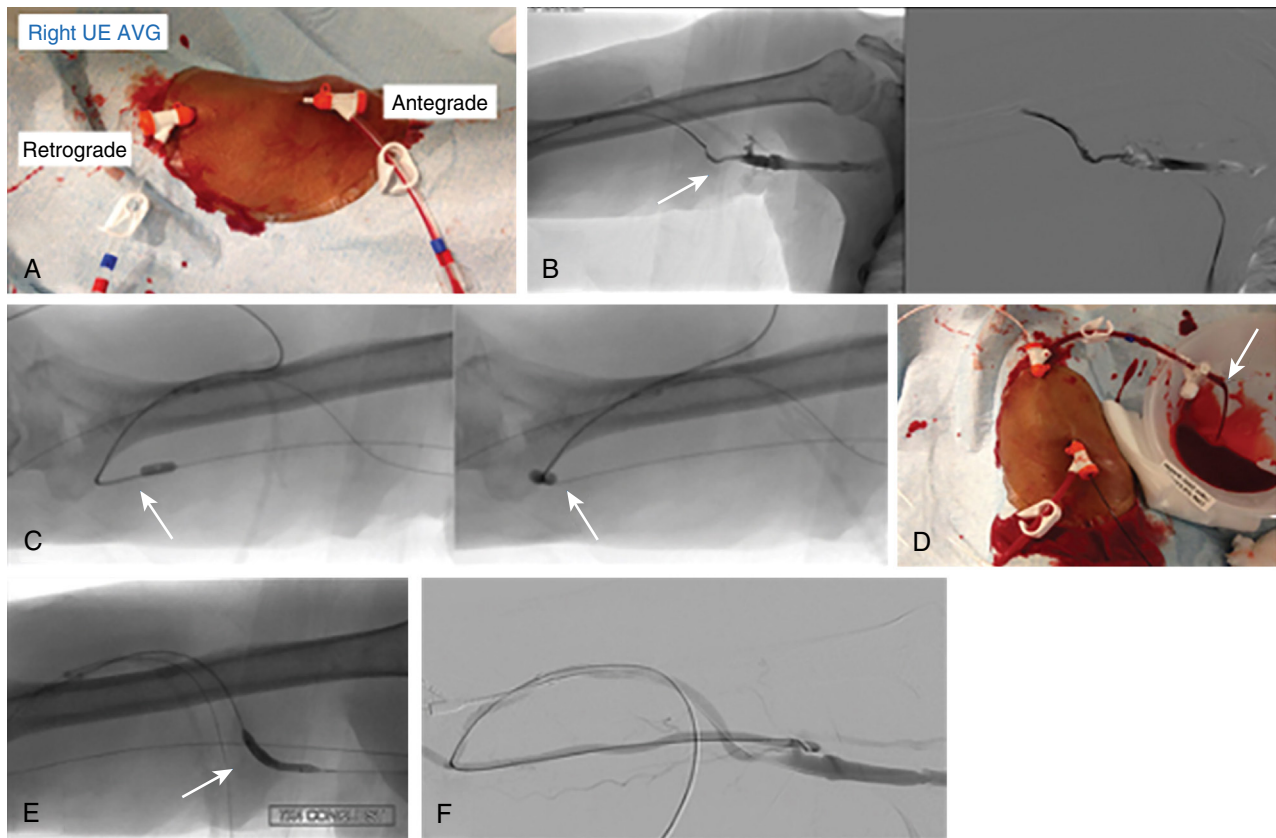


Fig. 67.20 Dialysis access thrombectomy using Fogarty balloon. (A) Two 7-French sheaths inserted in the arterial and venous sides of the graft (retrograde and antegrade, respectively). (B) Pullback contrast injection to determine the level of stenosis. (C) Removal of arterial plug using Fogarty balloon (arrow). (D) The arterial limb is patent (arrow—blood flowing from the sheath sidearm). (E) Angioplasty of the venous anastomosis (arrow—angioplasty balloon). (F) Final angiogram of the dialysis access showing patent flow. (From Almeih A, Sheta M, Abaza M, et al. Endovascular management of thrombosed dialysis vascular circuits. *Semin Intervent Radiol*. 2022;39[1]:14–22.)

USE OF STENTS IN DIALYSIS ACCESS

Balloon angioplasty is often associated with a poor patency rate necessitating repeated angioplasty. Endoluminal stents have been developed to improve the overall dialysis access functionality.¹⁷² Endovascular stents are scaffolds or tubelike structures that provide mechanical endoluminal support to the vessel wall to maintain patency. These include bare metal stents, nitinol stents, and stent grafts. In clinical practice, most of the used stents are stent grafts (covered stents), which are metal stents by design and are covered with a fabric coat such as Dacron or expanded polytetrafluorethylene (ePTFE) graft material. This design is intended to maintain vessel patency through the endoluminal wall support, while the biocompatible barrier is meant to prevent the tissue in-growth and restenosis.¹⁷² Several types of endovascular stents that have different properties are available for use in dialysis accesses.¹⁷²

Stents are used during access thrombectomy to prevent lesion recoil, vessel rupture, and recurrent stenosis.^{47,172} According to KDOQI 2019, the main indications for stent use in dialysis access are recurrent clinically significant graft-vein anastomotic stenosis in AVG; recurrent significant graft-vein anastomotic thrombosis in AVG; in-stent restenosis in AVF and AVG; treatment of ruptured venous stenotic segment of AVF and AVG; and management

of highly selected AV access aneurysms/pseudoaneurysms (not amenable to surgery).²⁴

Several studies reported the outcomes of stent use for refractory vascular access stenosis.¹⁷³⁻¹⁷⁹ However, most of them have been limited by their retrospective nature, absence of suitable control groups, lumping together of patent and thrombosed accesses, and combining grafts with fistulas.¹⁸⁰ For instance, one small randomized study compared the use of stents with conventional angioplasty and found no difference in primary AV graft patency after the intervention.¹⁷⁴ However, this study enrolled a mixture of clotted and patent AV grafts, and the stenotic lesions were at a variety of locations, thus limiting the interpretation of the findings.

Since primary AV graft patency is particularly short-lived in clotted AV grafts undergoing thrombectomy, those AV grafts may experience better patency after stent deployment. One series reported the outcomes of 34 clotted AV grafts undergoing thrombectomy with stent placement at the venous anastomosis.¹⁷⁷ The primary patency post intervention in this homogeneous group was 63% at 6 months. Although there was no matched control group treated with angioplasty alone, the primary AV graft patency was much higher than that reported previously (11%–39% at 6 months). A nonrandomized study comparing outcomes of clotted AV grafts treated with thrombectomy and stent

placement at the venous anastomosis with matched control patients treated with only thrombectomy and angioplasty observed a significantly longer primary patency in AV grafts treated with a stent, as compared with those treated with angioplasty alone.¹⁸¹ Similarly, a retrospective study with 48 patients observed longer AV graft patency after elective treatment of stenosis with stent deployment, as compared with angioplasty alone.¹⁸²

Several randomized controlled trials have evaluated the effects of stents on the vascular access patency among dialysis patients. The FLAIR Pivotal Trial compared the FLAIR stent with the standard PTA in managing stenotic lesions at the vein anastomosis of AVG. At 6 months, the primary patency of the target lesion was significantly better in the stent graft group (51% vs. 23%, $P < 0.001$).¹¹⁰ The REVISE trial compared the efficacy of the Viabahn stent grafts with angioplasty alone for the treatment of graft-vein anastomosis stenosis in both failing AVGs and thrombosed AVGs. This was the first randomized controlled trial to test the efficacy of stent grafts in thrombosed AVGs.¹⁸³ The primary patency of the target lesion at 6 months was better in the stent group as compared with the angioplasty group (51.6% vs. 34.2%, $P = 0.006$). Primary patency of the target lesion was higher in the dysfunctional AVGs compared with the thrombosed AVGs: 64.6% versus 36.1% in stent graft group and 45.8% versus 23.5% in the angioplasty group.¹⁸³ The RENOVA trial compared the efficacy of the FLAIR stent graft to angioplasty for managing the graft-vein anastomosis stenosis in AVGs. At 2 years, the primary patency of the target lesion was 26.9% and 13.5% ($P = 0.001$) in the stent graft group and angioplasty, respectively.^{172,184}

In-stent restenosis is a common complication of dialysis access stenting. This complication is routinely managed by standard balloon angioplasty. The use of stenting to treat in-stent stenosis was evaluated in the RESCUE trial, which randomized 275 patients with in-stent restenosis to either the stent graft group ($N = 132$) or PTA-alone group ($N = 143$). The authors reported that at 6 months, the access circuit primary patency was better in the stent graft group as compared with the angioplasty group: 18.6% versus 4.5% ($P < 0.001$).¹⁸⁵

Endovascular stents can also be used in special clinical scenarios, such as cephalic arch stenotic lesions that are resistant to angioplasty and stenotic lesions across the elbow, especially with vessel rupture.^{172,186}

PERCUTANEOUS ENDOVASCULAR ARTERIOVENOUS FISTULA (EndoAVF) FOR HEMODIALYSIS ACCESS

It is well acknowledged that AVF is the preferred dialysis access over both AVG and CVC due to its low infection rate and low intervention rate once matured.¹⁸⁷ Surgical AVF procedures have evolved over the past few decades using different locations and vascular territories in both the upper arm and forearm sites. Moreover, several techniques have been developed to improve the overall patency including creating the anastomosis between the perforating vein and brachial artery or proximal radial artery.¹⁸⁸ Importantly, proximal AVFs creation can be constructed in patients with small forearm veins and are rarely associated with high

blood flow through the AVF.¹⁸⁸ Nonetheless, surgically constructed AVFs are often plagued by the high primary failure rate ranging from 20% to 70%, which is often attributed to neointimal hyperplasia. The fundamental premise of endovascular creation of AVF (EndoAVF) is to minimize neointimal hyperplasia by reducing the vessel trauma encountered during surgery and subsequently improve the overall AV fistula maturation.

EndoAVF CONCEPT

Both the superficial and deep venous systems of the upper arm communicate via one or more perforator veins at the elbow level. Using the available EndoAVF devices, the perforator vein is utilized to divert the blood flow from deep to the superficial veins (cephalic and basilic), leading to the enlargement and arterialization of these veins, which eventually become suitable for cannulation.¹⁸⁸

The WavelinQ device (Becton, Dickinson and Company, Franklin Lakes, NJ) is a dual-catheter system, which deploys parallel catheters in adjacent artery and vein. The fistula is created between proximal radial or ulnar artery and the planned medial/lateral radial or ulnar vein in a side-to-side fashion. Once the fistula is created, the blood flows from the artery through the created anastomosis, toward the perforator, and then drains via the upper arm superficial veins (cephalic and/or basilic), as well as through the upper arm deep veins.^{188,189} The Ellipsys (Medtronic, Minneapolis, MN) device is a single-catheter system, which creates a direct anastomosis between the proximal radial artery and perforator vein in a side-to-side fashion. Angioplasty of the freshly created anastomosis is often performed. Blood flows from the artery through the anastomosis via the perforator vein, into the upper arm superficial veins (Fig. 67.21).¹⁹⁰ Most of these dialysis accesses require further procedures (banding, ligation, anastomosis angioplasty) before they become ready for cannulation.

CLINICAL OUTCOMES OF EndoAVF

Early results from a prospective, single-arm, multicenter study (Novel Endovascular Access Trial (NEAT) in 80 patients have demonstrated 1. successful endovascular AV fistula creation in 98% of participants; 2. 87% physiologic suitability for dialysis; 3. EndoAVF functional usability of 64% in participants who received dialysis; and 4. 12-month primary and cumulative patency of 69% and 84%, respectively.¹⁹¹ A multicenter, multioperator trial assessed the efficacy and safety of the Ellipsys device in a total of 107 patients and reported a technical success of 95% and a mean time to 2-needle dialysis of 100 days.¹⁹⁰ Multiple other clinical trials using these devices showed further improvement of the cumulative rates of EndoAVF-based dialysis circuits.

When compared with surgically created AVFs, Inston and colleagues¹⁹² found that outcomes in the WavelinQ EndoAVF group were comparable with the surgical group for both primary patency and cumulative patency rates, as well as the time to dialysis. On the other hand, Harika and colleagues compared Ellipsys EndoAVF with surgical fistulas and found that the EndoAVF group had a higher rate of maturation at 6 weeks (65% vs. 50%, $P = 0.02$) but a lower primary patency rate (61% vs. 86%, $P = 0.01$), as well as a higher rate of intervention (50% vs. 34%, $P = 0.013$) at 1 year.¹⁹³

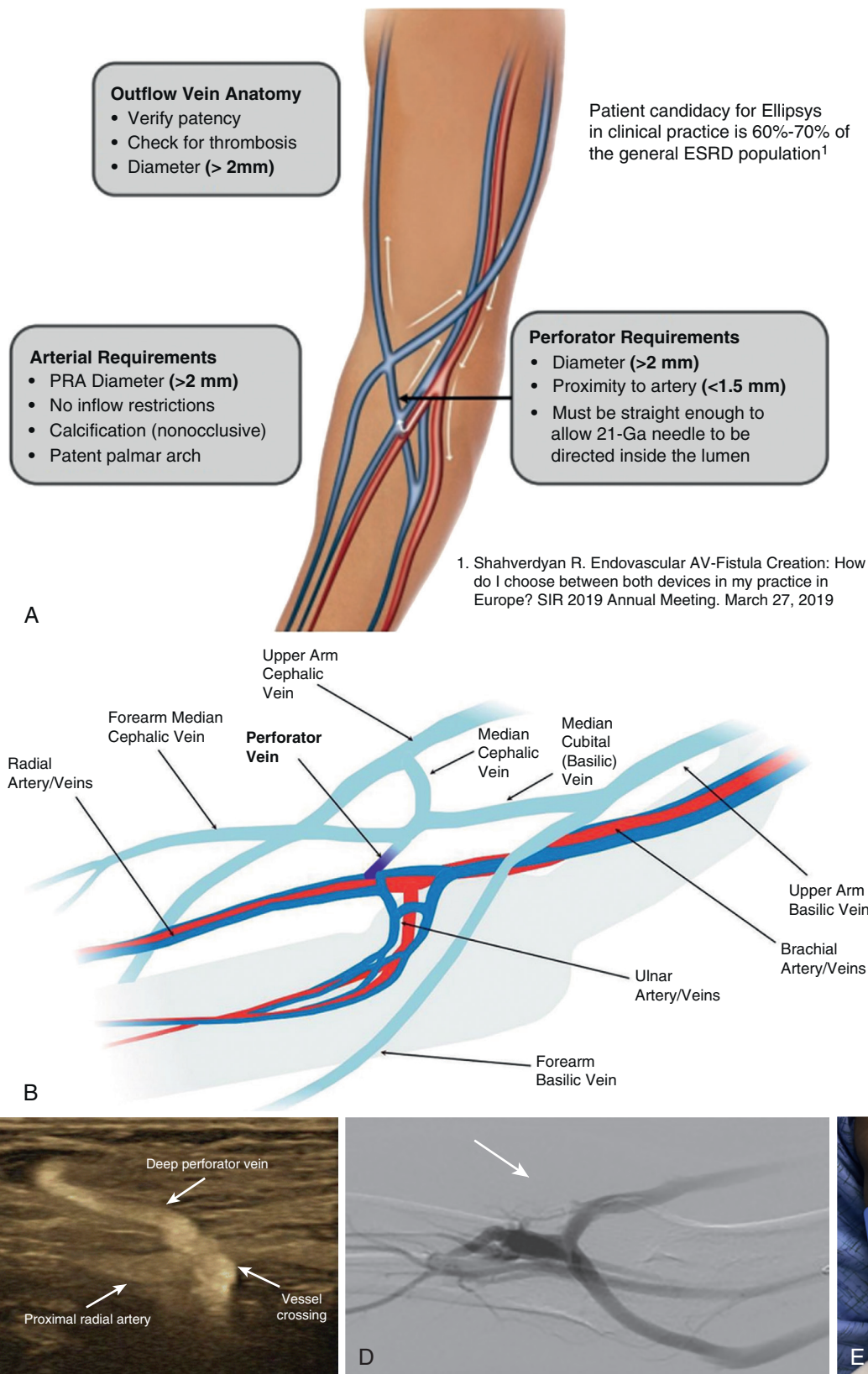


Fig. 67.21 (A) Criteria of vein and artery size. (B) Blood vessel supply of forearm. (C) Ultrasonography showing both the perforator vein and radial artery. (D) Angiogram showing fistula creation between the perforator and proximal radial artery. (E) Successful cannulation of the fistula. (A from "Updated Mapping Protocols for Percutaneous AV Fistula Creation Sara Skjonsberg BS, RVT, RPhS" EnVveno Medical Corporation Bend, Oregon <https://strandness.org/wp-content/uploads/2024/02/26-AM0820-SKJONSBERG-SARA.pdf>. B from Shahverdyan R, Konner K, Segerer S, Wasse H. Novel classification of proximal forearm perforator vein in the planning and creation of percutaneous and surgical Gracz-type arteriovenous fistulae. *J Vasc Access*. 2024;25[3]:872–882. C from Isaak A, Jörg L, Attigah N, et al. Practical guide of vascular ultrasound in arteriovenous fistulae. *Vasa*. 2023;52[1]:22–28. E from Kumbar L, Ramani K, Brouwer-Maier D. Considerations in access cannulation: traditional and evolving approaches. *Adv Chronic Kidney Dis*. 2020;27[3]:199–207.)



Fig. 67.22 Digital subtraction angiography showing a severe stenosis of the left innominate vein. There are multiple ipsilateral and across the neck collaterals draining into a normal right innominate vein.

This variation in outcomes may reflect the initial learning curve and heterogeneity of the surgical AVF population.

Taken together, while current EndoAVF data are promising, there is a need for head-to-head comparison with surgical AVFs. Further, educating dialysis staff on cannulating EndoAVF is a major challenge. Finally, the long-term complications of EndoAVF fistulas are not known, such as steal syndrome, pseudoaneurysm formation, and high-output heart failure.¹⁹⁴

CENTRAL VEIN STENOSIS

Central vein stenosis (CVS) is a frequent occurrence in patients on hemodialysis.^{29,195,196} Acute or chronic trauma of the central vessels caused by either temporary or permanent dialysis catheters is the major cause, possibly exacerbated by catheter infections.¹⁹⁷ Central vein stenosis leads to impairment of venous return on the ipsilateral extremity and may, in turn, result in malfunction or thrombosis of the vascular access.¹⁹⁸ Although it may be asymptomatic, patients with central vein stenosis most commonly present with ipsilateral upper extremity edema. In some patients, a previously unappreciated CVS becomes evident clinically following the creation of an ipsilateral AV fistula or graft. The diagnosis can be confirmed by angiography, ultrasound, or magnetic resonance venography.

The most commonly encountered location of CVS is at the junction of the cephalic vein with the subclavian vein (not catheter injury related). Other central veins that may be affected (often related to injury from previous catheters) include the subclavian vein, brachiocephalic vein, and superior vena cava (Fig. 67.22). In patients with tunneled femoral catheters, central vein stenosis may occur in the external iliac

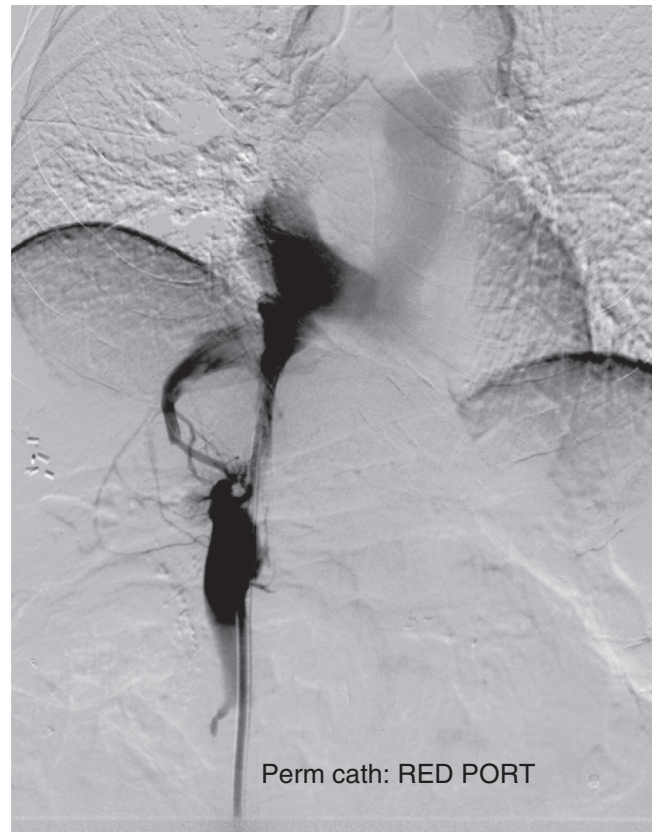


Fig. 67.23 Digital subtraction angiography showing a severe stenosis of the infrahepatic portion of the inferior vena cava with some intrahepatic collaterals.

vein, common iliac vein, or inferior vena cava (Fig. 67.23), resulting in ipsilateral lower extremity edema. The stenotic lesion results from neointimal proliferation, or clot and fibrin sheath formation around an indwelling dialysis catheter that becomes organized and incorporated into the vessel wall. Stenoses may progress over time to complete occlusion of the venous circulation (Fig. 67.24). If left untreated, central vein stenosis is often associated with increased retrograde pressure and formation of venous collaterals (Fig. 67.25). In some patients, the collaterals are sufficiently well developed to permit adequate venous drainage that prevents formation of edema (Fig. 67.26A and 67.26B).

Balloon angioplasty is the standard treatment for symptomatic central vein stenosis.¹⁹⁹⁻²⁰⁶ The long-term success of PTA of CVS is quite poor due to a combination of elastic recoil and aggressive neointimal hyperplasia. In one study, the primary patency was substantially shorter after angioplasty of central vein stenosis, as compared with stenoses at more peripheral locations.⁷⁸ As a result, patients with CVS may require multiple angioplasties to treat recurrent lesions.

Stent placement has been attempted in the management of refractory central vein stenosis due to elastic recoil (Fig. 67.27). Several small series have reported the outcomes of stent placement for refractory central venous stenotic lesions. These studies have been limited by their retrospective study design, the small numbers of patients, and the absence of a control group. In two uncontrolled series, the primary patency following stent deployment

for central vein stenosis was 42% to 50% at 6 months and only 14% to 17% at 1 year.^{199,202} A metaanalysis of 8 studies that evaluated 473 patients concluded that angioplasty may provide a better primary assisted patency rate as compared with stenting.²⁰⁷ In patients with ipsilateral vascular access and persistent upper extremity edema, which is resistant to angioplasty, the only effective options could be the ligation of the vascular access and creation of a contralateral arteriovenous access, transition to peritoneal dialysis, or urgent kidney transplantation.

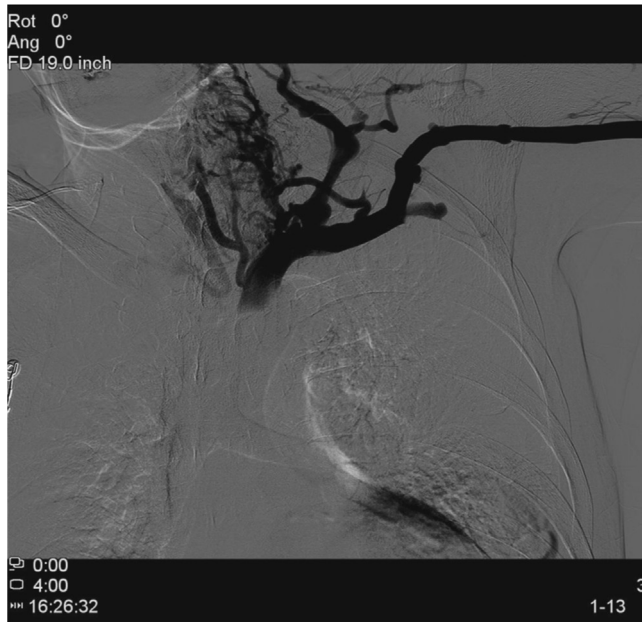


Fig. 67.24 Digital subtraction angiography showing a complete occlusion of the left innominate vein.

ACCESS-INDUCED HAND ISCHEMIA

Clinical manifestations of access-induced hand ischemia can vary widely, ranging from mild (hand pain) to severe (tissue necrosis with loss of digits). Vascular accesses have a normal retrograde blood flow, which does not cause hand ischemia. There are three recognized mechanisms for access-induced hand ischemia: 1. true arterial steal (with complete retrograde blood flow toward the graft or fistula), 2. arterial stenosis distal to the arterial anastomosis, and 3. generalized arterial calcification (most commonly seen in patients with diabetes mellitus) with distal digital occlusions (Fig. 67.28). Since all three mechanisms may coexist in a single patient, “distal hypoperfusion ischemic syndrome” is the preferred term.²⁰⁸ Other causes of hand pain, such as carpal tunnel syndrome, arthropathies, diabetic neuropathy, ischemic monomelic neuropathy, and reflex sympathetic dystrophy syndrome, should be excluded before making a definitive diagnosis of vascular access-induced hand ischemia.

Physical examination is the most important tool to diagnose access-induced hand ischemia. Pale, cold digits during and between hemodialysis sessions are seen in almost 90% of the cases. Additional diagnostic tools include digital brachial index measurement, digital plethysmography, duplex ultrasound, and transcutaneous oxygen saturation. If these noninvasive tests are suggestive of access-induced hand ischemia, then a complete angiogram of the vascular access, including the feeding artery, should be performed.

Management of access-induced hand ischemia is directed at alleviating the underlying etiology. Treatment options range from angioplasty of the stenosis distal to the anastomosis to ligation of the vascular access. Other interventions that have been successful in relieving the ischemia while maintaining patency of the AV fistula

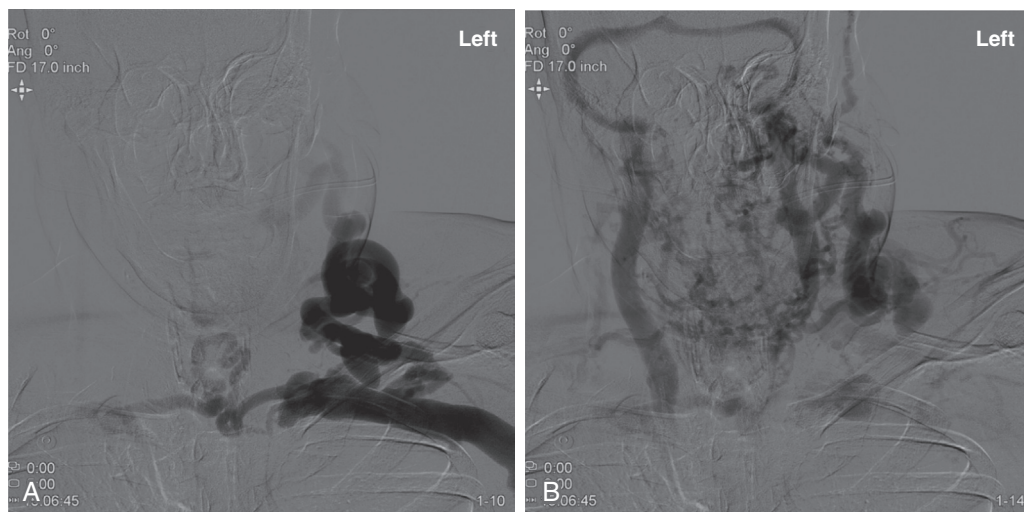


Fig. 67.25 (A) Digital subtraction angiography (DSA) showing a complete occlusion of the left innominate vein with severe enlargement of the left internal jugular and multiple ipsilateral and across the neck collaterals draining into a normal right innominate vein. (B) DSA of the same patient showing the drainage of the left internal jugular to the right internal jugular across the cavernous veins.



Fig. 67.26 (A) Digital subtraction angiography (DSA) showing a complete occlusion of the superior vena cava with severe augmentation of the azygous vein. (B) DSA of the same patient showing the drainage of the azygous vein into the lumbar and intercostal vein.

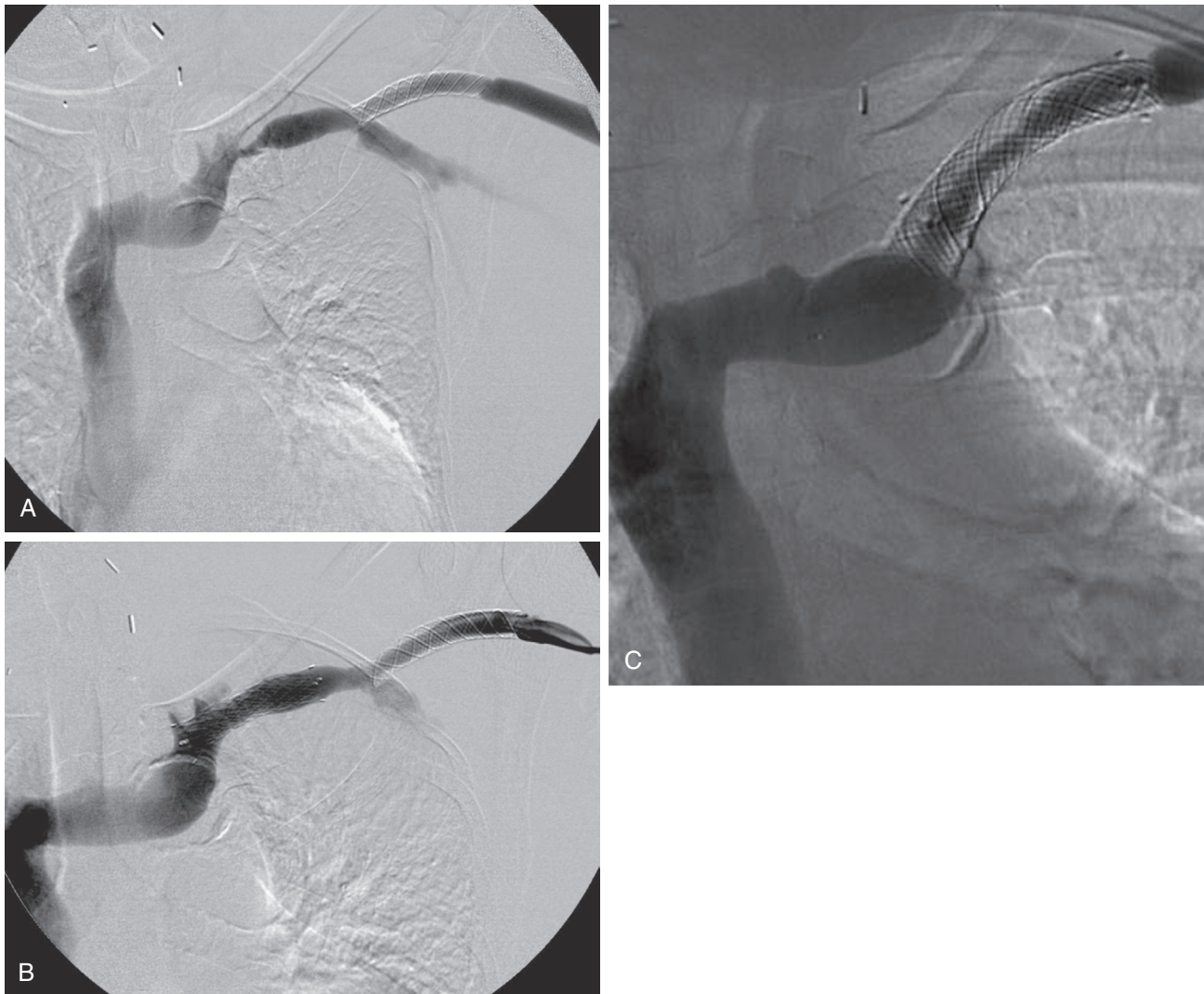


Fig. 67.27 (A) Digital subtraction angiography (DSA) showing a severe stenotic lesion of the left subclavian. There is also a stent in the left cephalic vein. (B) DSA showing a stent deployed at the severe stenotic lesion of the left subclavian with excellent initial results. (C) DSA taken 12 months after the initial placement of a stent in the left subclavian showing intra-stent stenosis due to significant myointimal hyperplasia.

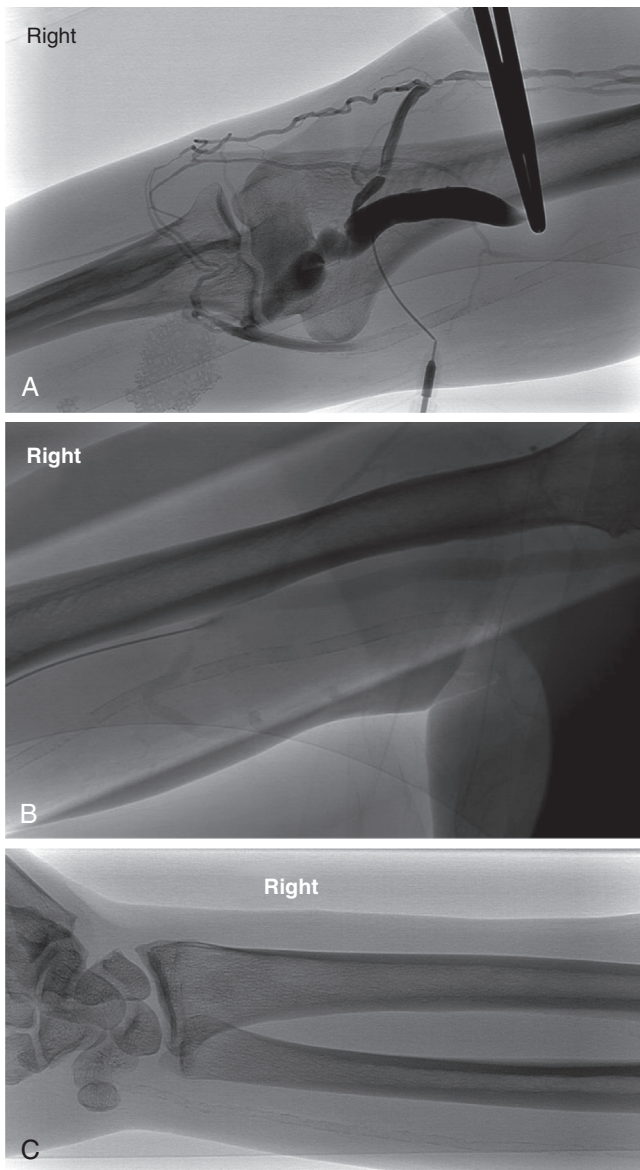


Fig. 67.28 (A) Angiogram of a right upper arm fistula demonstrating an arterial steal syndrome on a patient with generalized arterial disease. (B) Spot film of the same patient showing a calcified brachial artery. (C) Spot film of the same patient showing a calcified ulnar artery.

include banding or plication of the AV fistula (Fig. 67.29), graft interposition, and distal revascularization with interval ligation procedure.²⁰⁹

HEMODIALYSIS CATHETERS

NONTUNNELED TEMPORARY HEMODIALYSIS CATHETERS

Temporary hemodialysis catheters are indicated for acute dialysis treatments. These catheters are made of polyurethane, polyethylene, or polytetrafluoroethylene; they have a double lumen, are semirigid, and are relatively easy to place in the internal jugular (preferably on the right side), femoral, and rarely subclavian veins. According to



Fig. 67.29 Access to the right internal jugular vein guided by real-time ultrasound.

KDOQI 2019 guidelines, it is recommended to limit the use of nontunneled catheters to a maximum of 2 weeks due to the high risk of infection.^{24,210} Importantly, the subclavian vein is usually used only if there is no other access available; this is because of its higher risk of pneumothorax and increased risk for central venous stenosis that can compromise the creation or placement of future AV fistulas or grafts on the same side.¹⁹⁷ It is recommended that the tip of the catheter be at the junction of the right atrium and superior vena cava for chest catheters and within the inferior vena cava for femoral location. Temporary hemodialysis catheters can be placed blindly by ultrasound or fluoroscopic guidance. The use of real-time ultrasound decreases the number of attempts at vein cannulation and minimizes the risk of inadvertent arterial cannulation. A chest radiograph should always be obtained immediately following placement of these catheters before hemodialysis is initiated to confirm line placement and exclude pneumothorax. Radiograph imaging is not required after the placement of a femoral dialysis catheter. Potential early complications include pneumothorax, vein or arterial perforation, mediastinal or pericardial perforation with the possibility of hemothorax and cardiac tamponade, air embolism, and local hematoma with possible extension into the soft subcutaneous tissue of the neck and possible external obstruction of the airways. Long-term complications include development of stenotic lesions along the trajectory of the catheter, which may preclude the use of the ipsilateral limb for future creation of a vascular access. If the patient already has a documented stenotic lesion of the central vessels, placement of an indwelling catheter may cause a life-threatening acute central vessel occlusion. The complications at the femoral site are less dramatic, but vein or arterial perforation and formation of arteriovenous fistula are possible. Deep vein thrombosis (either related to the catheter itself, or immobilization related to prescribed bed rest), local hematomas, exit site infections, and bacteremia are other important complications. Exit site infections and catheter-related bacteremia are frequent complications of temporary dialysis catheters. Development of catheter-related bacteremia requires immediate institution of systemic antibiotics and removal of the nontunneled dialysis catheter.

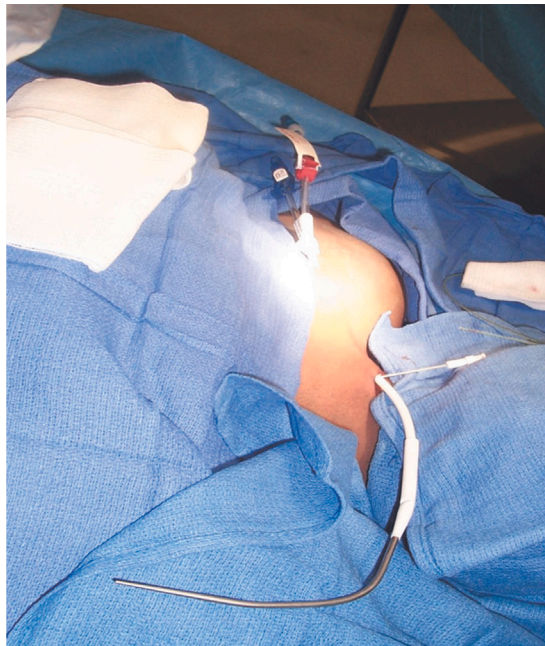


Fig. 67.30 The catheter has been tunneled from the upper part of the chest to the internal jugular site entrance.

TUNNELED HEMODIALYSIS CATHETERS

Tunneled hemodialysis catheters (THCs) are an essential part of kidney replacement therapy that are commonly used in ESKD patients. THCs are considered reasonable for providing chronic dialysis in certain clinical scenarios: All other options of permanent AV access in all four extremities have been exhausted; waiting for live-donor kidney transplant within 90 days; and temporary switch from another modality due to complication (PD infection, graft rejection); limited life expectancy; patient's choice; and clinical conditions that would worsen with AV access creation (advanced heart failure).²⁴ The right internal jugular vein is the preferred site due to short distance from the right atrium and its straight trajectory. In general, THCs have the same characteristics as the temporary catheters but are longer and have a Dacron cuff located within the tunneled portion of the catheter in the subcutaneous tissue. Following insertion, an inflammatory response around the cuff results in scar tissue, creating a mechanical barrier that prevents introduction of infection from the exit site into the bloodstream. Thus the frequency of catheter-related bacteremia is lower with tunneled dialysis catheters as compared with acute nontunneled catheters.^{211,212}

The insertion of THCs is usually performed using real-time ultrasound (Fig. 67.29) and fluoroscopy guidance. The tunnel is created inferiolaterally under the clavicle, approximately 5 to 7 cm from the initial needle puncture of the internal jugular vein at the base of the neck (Fig. 67.30). It is recommended that the tip of the catheter be at the junction of the superior vena cava and right atrium (Fig. 67.31) and the cuff should be 2 cm from the exit site. Radiography or fluoroscopy is required to assess for any pneumothorax and catheter kink after placement (Fig. 67.32).

Complications at the time of placement are similar to those associated with temporary catheters (Figs.

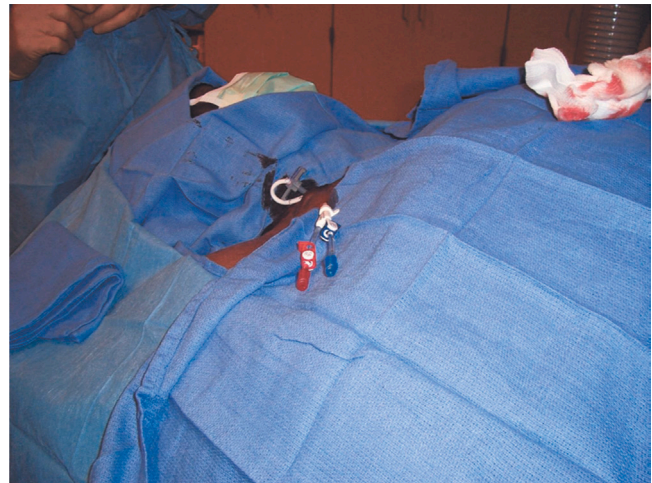


Fig. 67.31 The catheter is introduced into the opening of the peel-away sheath and fed up to the junction of the superior vena cava and right atrium.



Fig. 67.32 Radiograph of an appropriate placement of a right internal jugular vein tunneled chronic dialysis catheter.

67.33–67.36). Internal jugular thrombosis develops in about 25% of tunneled catheters but is usually asymptomatic.²¹³ With regards to removal, THCs can be safely removed at bedside under local anesthesia.²¹⁴

LESS COMMON LOCATIONS FOR TUNNELED HEMODIALYSIS CATHETERS

FEMORAL TUNNELED HEMODIALYSIS CATHETERS

The use of femoral vein location is indicated usually in cases of central venous occlusion that is usually related to the prolonged use of upper extremity THCs.^{215,216} Femoral THCs are usually longer than chest catheters (28–55 cm); the catheter tip is located in either the proximal inferior vena cava or right atrium (Fig. 67.37A and 67.37B).²¹⁵ The subcutaneous tunnel is created in the anterior upper thigh.

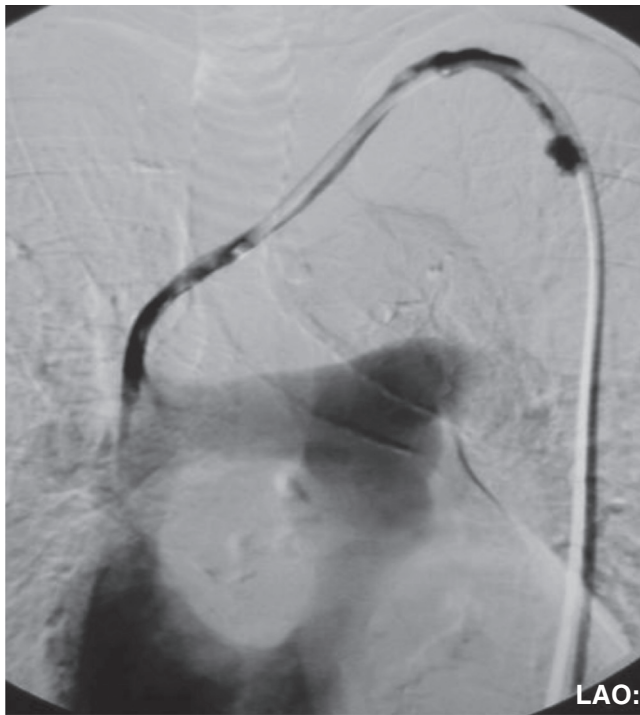


Fig. 67.33 Tunneled catheter complication. Fibrin sheath around the left internal jugular vein catheter.



Fig. 67.34 Tunneled catheter complication. The catheter was placed in the right carotid artery with the tip near the aortic valves. Contrast demonstrates the aortic arch.

Notably, the primary patency of tunneled femoral catheters is significantly worse than that of tunneled internal jugular catheters.²¹⁵ Presumably, some failures are due to kinking of the catheter in the groin when the thigh is flexed. The frequency of catheter-related bacteremia is similar for patients with femoral and internal jugular dialysis catheters. The likelihood of catheter-related bacteremia is proportional to the duration of catheter use.^{198,217} There is high (~25%) frequency of symptomatic ipsilateral deep vein thrombosis after placement of a tunneled femoral catheter.²¹⁵ Fortunately, this complication can be treated with long-term anticoagulation, thereby permitting continued use of the catheter.

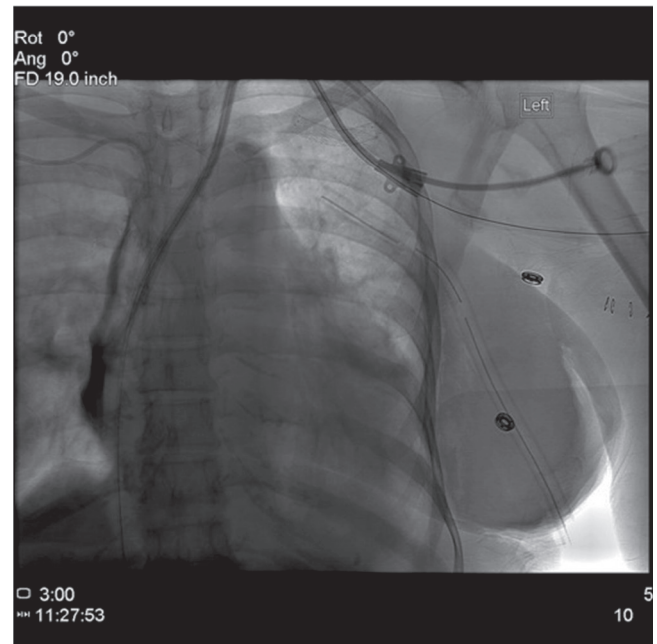


Fig. 67.35 Tunneled catheter complication. Left internal jugular vein catheter with its tip located in the pericardial sac. Contrast demonstrates the outer lining filling defect that corresponds to the lower part of the adventitia of the superior vena cava and upper part of the pericardium.

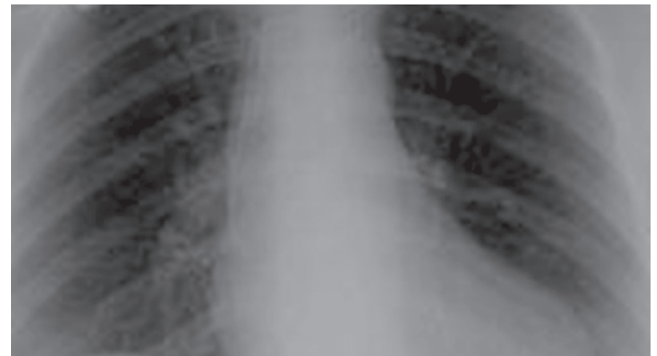


Fig. 67.36 Tunneled catheter complication. Foreign body (metallic wire) that was left behind after an attempt for an internal jugular vein catheter placement.

TRANSLUMBAR AND TRANHEPATIC CATHETERS

Translumbar and transhepatic THC's are considered the last resort in dialysis patients who exhausted all other access options.

For *translumbar catheters*, there are essentially two tunnels. One tunnel extends from the access site in the back toward the IVC and the second from the abdomen to the access site in the back (Fig. 67.38).^{218,219}

On the other hand, the placement of tunneled translumbar hemodialysis catheters requires greater expertise in interventional techniques than jugular or femoral access, especially since ultrasound guidance is not possible. The most common complications of translumbar catheters are the partial dislodgment and bleeding with subsequent retroperitoneal hematoma.²¹⁹

Regarding *transhepatic THC's*, these catheters are placed using the right upper quadrant using ultrasound and fluoroscopy guidance. In brief, a 21-gauge needle is placed

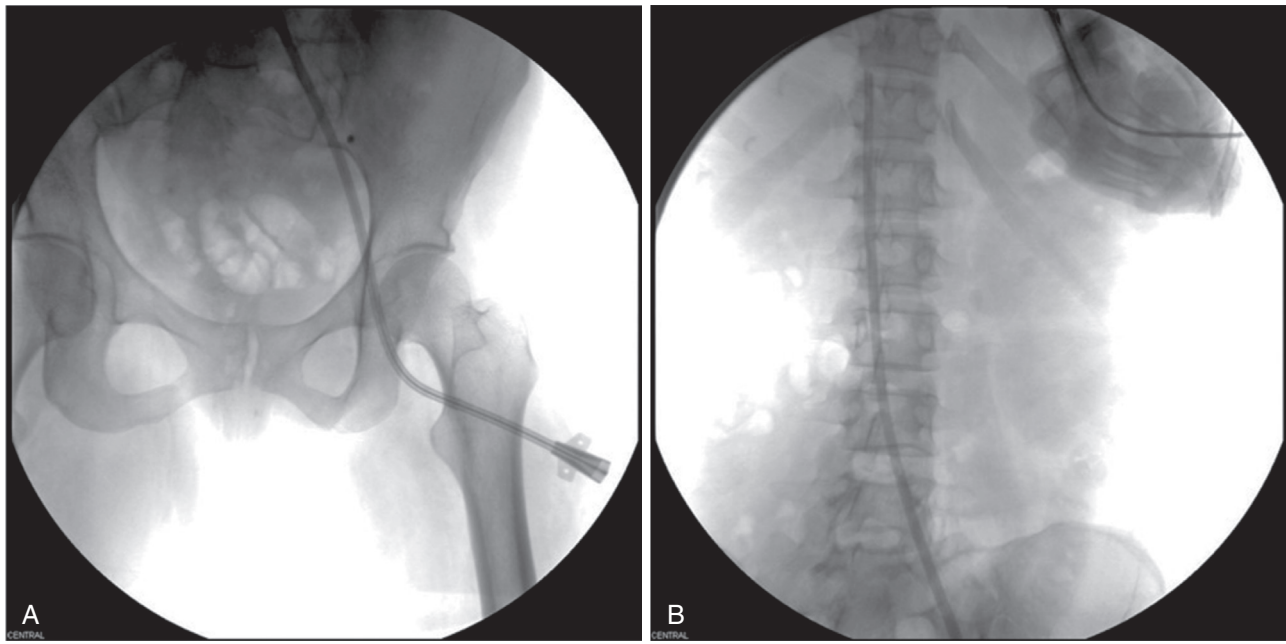


Fig. 67.37 (A and B) X-ray films of an appropriate placement of a left femoral vein tunneled chronic dialysis catheter.

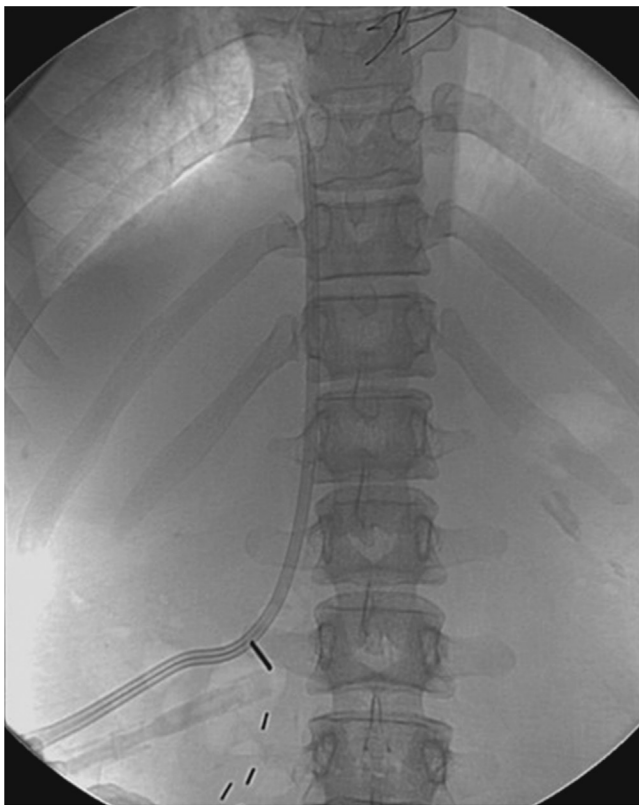


Fig. 67.38 X-ray film of an appropriate placement of a translumbar tunneled chronic dialysis catheter.

halfway through the liver in a direction parallel to the right of the middle hepatic veins under fluoroscopy guidance; then, contrast material is injected through the needle and the needle withdrawn until a hepatic vein is visualized. Once a suitable vein is accessed, a guidewire is placed and advanced to the right atrium. A subcutaneous tunnel is created inferiorly to the insertion site and a dual-lumen

cuffed hemodialysis catheter is placed.²²⁰⁻²²² The major complications are bleeding, perihepatic hematoma, hepatic vein thrombosis, and the dislodgment of the catheter due to liver excursion movement with breathing.²²⁰⁻²²²

COMPLICATIONS OF TUNNELED HEMODIALYSIS CATHETERS

Catheter dysfunction is diagnosed when blood cannot be aspirated from the catheter lumen at the time of dialysis initiation or, more commonly, if it is not possible to consistently achieve a blood flow required to yield sufficient urea and other solute clearance (>250 mL/min). In catheters that were previously delivering an adequate blood flow, intraluminal thrombus is the most common etiology for dysfunction, although a fibrin sheath may be the culprit in some patients. The management of THC dysfunction related to thrombosis includes the administration of 2 mg of tPA with 4% citrate in each port using dwell or push method with a dwell time of at least 30 minutes.²⁴ If the thrombolytic agent fails to restore the flow, the presence of fibrin sheath is suspected and the THC usually requires exchange. The formation of a fibrin sheath is a common cause of THC dysfunction, affecting 40% to 100% of patients. In most centers, contrast injection during THC exchange establishes the diagnosis of fibrin sheath and catheter exchange alone or, most commonly, in combination with balloon angioplasty is used to treat the fibrin sheath.²⁴ To prevent THC dysfunction, low-concentration citrate (5%) or heparin can be used for locking the catheter. The use of tPA prophylactically once weekly for locking THC is also a reasonable approach to address this complication.²⁴

Catheter-related infections are the most common complications that are associated with increased morbidity, mortality, and health care costs. Catheter exit site infection presents with the presence of drainage and systemic signs and symptoms that requires cultures from both exit site

and blood cultures taken from the catheter, followed by empiric antibiotics that cover gram-positive organisms for 7 to 14 days. Catheter removal is rarely indicated in exit site infection unless there is a tunnel infection.²²³ Exit site infections can be prevented with careful exit site care (see Chapter 62). Tunnel infections present with redness over the catheter tunnel, likely concomitant exit site infection, and possible fluid collection along the tunnel, visible on ultrasound. Cultures must be obtained from the drainage and blood cultures from catheter. Empiric antibiotics should cover both gram-positive and gram-negative organisms initially and then be tailored to culture results and continued for 10 to 14 days. If these measures fail to eradicate the infection, then the catheter needs to be removed and a new one should be inserted at another entry site.

Catheter-related bloodstream infection (CRBSI) is a common indication for catheter replacement.²⁸ CRBSI is defined as the presence of clinical manifestations and at least 1 positive blood culture obtained from a peripheral source (vein or dialysis circuit) and no other apparent source, with positive semiquantitative (>15 CFU/catheter segment, hub or tip) or quantitative ($>10^2$ CFU/catheter segment, hub or tip) culture whereby the organism is isolated from the catheter and a peripheral source.^{24,223} The clinical manifestations of CRBSIs include fever, chills, rigors, and headache. The clinical management of CRBSIs currently involves an individualized approach that takes into consideration the overall patient's health, vascular access history, and dialysis needs.²²³ When CRBSIs are suspected, blood cultures from the catheter and a peripheral source should be obtained and empiric antibiotics should cover both gram-negative and gram-positive organisms, potentially including coverage for methicillin-resistant *Staphylococcus aureus* depending on local sensitivities. Antibiotics should be narrowed as soon as the culture results and sensitivities are available. The duration of antibiotics ranges from 10 to 14 days for gram-negative bacilli, 14 days for *Candida*, and 4 to 6 weeks for noncomplicated *S. aureus*.^{24,223} The removal of THC is indicated in the following situations: clinical and hemodynamic instability; persistent fever after 48 to 72 hours of antibiotic initiation; persistent bacteremia after 48 to 72 hours of antibiotic initiation; *S. aureus*, *Pseudomonas aeruginosa*, fungi or mycobacteria infection; and metastatic complications (e.g., endocarditis, epidural abscess).²²³

A thrombus inside the catheter may act as a nidus for the catheter biofilm. Heparin coating of the catheters may prevent bacterial adherence.²²⁴ Two randomized clinical trials evaluated short-term, nontunneled central vein catheters found a lower rate of catheter-related bacteremia in patients with heparin-coated catheters than those with noncoated catheters.^{224,225} Fewer infections have also been observed in patients with heparin-coated tunneled dialysis catheters compared with those using uncoated catheters.²²⁶ Instillation of tPA instead of heparin into the dialysis catheter lumen once weekly, as compared with the standard use of heparin locks three times a week, was also found to significantly reduced the incidence of catheter malfunction and bacteremia,²²⁷ suggesting that tPA may prevent thrombosis and biofilm development. The use of selective prophylactic antibiotic locks is reserved for catheter-dependent patients who are at high risk for CRBSIs.²⁴

PERITONEAL DIALYSIS CATHETER INSERTION

Peritoneal dialysis (PD) is an attractive alternative to hemodialysis in patients with ESKD. The attractiveness to PD modality stems from its cost-effectiveness, low complication rate, and improved quality of life.²²⁸ In-depth discussion of peritoneal dialysis can be found in Chapter 63. PD catheters can be placed into the abdominal cavity by surgeons,²²⁹⁻²³¹ interventional radiologists,²³² or interventional nephrologists.²³³ Several techniques have been described to place PD catheters including blind (Seldinger) technique,²³⁴ surgical placement,²²⁹ peritoneoscopic,²³⁵ laparoscopic,²³⁶ Moncrief-Popovich technique,²³⁷ and fluoroscopic insertion.²³³

Peritoneal catheters are made of either silicone rubber or polyurethane. The Tenckhoff catheter is the most common type of PD catheter placed. The intraperitoneal portion of the catheter can be straight, coiled, Ash (T-Fluted), or with a silicone disc.²³⁸ The extraperitoneal portion of the catheter may be straight or have a swan-neck design with single or double inner cuffs, or a combination of a single cuff and a silicone disc. The most widely used PD catheter is the double cuff, swan neck, coiled Tenckhoff design. This design has been shown to decrease mechanical complications (i.e., inflow and outflow problems). The coiled catheter design also decreases pain during infusion and is less likely to migrate. The swan-neck design was introduced to avoid cuff extrusions.²³⁹ The intraperitoneal portion of the catheter should be placed between the visceral and parietal peritoneum near the pouch of Douglas. The inner cuff should be inserted in the abdominal wall musculature (rectus muscle) to prevent leaks. The outer cuff is usually located in the subcutaneous tissue to create a dead space in between the two cuffs, which it is believed prevents migration of infections coming from the exit site. The subcutaneous tract and exit site should face downward or laterally to avoid exit site infection. The exit site should be determined and marked before insertion while the patient is in the upright position. The beltline, prior surgical sites, and abdominal midline should be avoided. Postoperative catheter care is important. The catheter should be covered with a nonocclusive dressing and should not be used for 10 to 14 days in elective start PD. The catheter should be flushed at least two or three times per week with saline or dialysate solution until the patient is ready to start PD.²⁴⁰ In practice, the use of PD often starts between 2 and 4 weeks after catheter placement. The rationale for waiting 2 weeks is to provide a sufficient time to allow healing around the cuffs, which helps in preventing leaks and subsequently decreasing the risk of infection.

Due to the distinct advantages of PD, some clinicians advocate for urgent dialysis PD. The urgent-start PD is introduced in those patients who need immediate dialysis initiation but have no preestablished vascular or PD access. PD can be started within 24 hours of catheter insertion.²⁴³ In urgent-start PD, low-volume PD is recommended while the patient is in recumbent position.²⁴¹ PD catheter outcomes were evaluated in elective start and urgent start groups, and

the study found that clinical outcomes were comparable except for catheter leak, which was expectedly higher in urgent start group.²⁴²

Two studies comparing swan-neck and straight Tenckhoff catheters have shown a similar risk for peritonitis and exit infection but less cuff extrusion with the swan-neck design. The lower incidence of cuff extrusion enhances the survival of the swan-neck catheters.²⁴³⁻²⁴⁵ A modified technique, which places the swan-neck catheter in a presternal exit site location, has been reported by Twardowski and colleagues^{246,247} and has shown an increase in access survival up to 95% at 2 years. The presternal exit site modification has been shown to decrease the incidence of peritonitis and exit site infection. The authors advocate the use of the presternal catheters in obese patients, patients with ostomies, and children wearing diapers and/or with fecal incontinence.^{246,247} Gadallah and colleagues²⁴⁸ demonstrated that placement of PD catheters by peritoneoscopic approach had a longer survival rate than those placed surgically and that the rates of exit infection and leak were lower. Moncrief and colleagues²³⁷ described a technique in which the extraperitoneal portion of the catheter is buried in the abdominal subcutaneous tissue until the patient is ready for PD. It appears that the Moncrief modification may lower the risk of initial infection of the tract.²³⁷

In real-world clinical practice, whereas laparoscopic PD placement is mostly performed by surgeons, percutaneous image-guided placement using ultrasound guidance is the method preferred by interventionalists. Comparative studies of the percutaneous versus open or laparoscopic surgical approaches are limited. The only randomized trial was performed in Iran, where 64 patients were randomized to either the percutaneous or open surgical approach. The main significant difference in outcome was increased rates of hemoperitoneum and outflow failure in the open surgical approach.²⁴⁹ All other studies are case series that are limited by the fact that open surgical cases are more often performed in patients who have had previous surgeries, a finding that may impact clinical outcomes, especially among those who have significant adhesions resulting in poor flow. It is reassuring that the percutaneous approach appears at least as effective as the open surgical approach. A retrospective study from Thailand of 205 patients (56 percutaneous) showed no difference in mechanical complications but a higher peritonitis rate in the open surgical approach.²⁵⁰ The largest study compared 283 percutaneous with 150 open surgical PD catheters and reported 21% versus 23% poor drainage, 6% versus 10% leak, and 4% versus 13% peritonitis, respectively.²⁵¹ A meta-analysis of 2681 patients showed no significant difference in catheter survival or dysfunction but an increased peritonitis rate in surgically placed catheters.²⁵² The International Society for Peritoneal Dialysis guidelines for creating and maintaining optimal PD access supports the use of all three techniques and recommends that centers should concentrate on techniques they are familiar with as this results in best outcomes.²⁵³

The major complications encountered during surgical PD insertion include bowel perforation and inferior epigastric artery injury. Reducing the risk of bowel perforation due to adhesions can be achieved using

abdominal ultrasound and assessing for “visceral slide,” a technique that uses a linear ultrasound probe placed longitudinally on the abdomen at the site of insertion. The patient is asked to breathe, and the bowel movement is measured. Shanmugalingam and colleagues²⁵⁴ used a slide of <1 cm to distinguish those who required an open approach; however, other studies in laparoscopic surgery have suggested 2 cm for safety. Complications of the percutaneous approach include bowel perforation, hemoperitoneum, and peritonitis. The former is rare and seen in <1% of catheter insertion when auditing against the ISPD guideline standard.²⁵³ Treatment of bowel perforation requires bowel rest, intravenous or oral antibiotic therapy, and rarely surgical exploration.^{229,255} Bleeding from the inferior epigastric artery or its branches due to inadvertent injury can manifest clinically with hypotension, tachycardia, abdominal pain, and severe drop in hemoglobin. In most cases, this complication is usually treated conservatively and rarely requires transarterial embolization.^{242,256}

PERITONEAL CATHETER PLACEMENT USING FLUOROSCOPIC AND ULTRASOUND TECHNIQUE

The patient is examined in detail and the belt line is determined in both sitting and standing positions. Both the entry and exit sites for the catheter are marked using a nonsterile double-cuff PD catheter as a template. Next, the upper border of the PD catheter curl is aligned to the upper border of the symphysis pubis while the patient is in the supine position. The entry site location, usually 2 to 3 cm lateral to the midline, is marked to prevent injury of the inferior epigastric artery. The exit site should be at least 2 to 4 cm away from the superficial subcutaneous cuff.²⁵⁷ The urinary bladder should be emptied before catheter insertion, or a Foley catheter can be placed if clinically needed. Preprocedural prophylactic antibiotics are usually administered such as intravenous cefazolin or vancomycin in case of allergy.²²⁸

TECHNICAL PROCEDURE

The PD catheter placement is performed under conscious sedation while the patient is in supine position.²²⁸ Under ultrasound guidance, a 21-gauge needle is used to access the peritoneal space. Then, 3 to 5 mL of radiocontrast are injected into the peritoneal cavity under fluoroscopy to ensure appropriate location: A radiologic pattern of outer bowel delineation is indicative of a good placement. Next, a 0.018-inch guidewire (Glidewire Guidewire 0.035" Dia. x 180 cm L, Terumo, Somerset, NJ) is introduced through the needle and then exchanged for a 6-French catheter sheath. A 2- to 4-cm incision is made on the skin, and the subcutaneous tissue is digitally dissected up to the rectus muscle. A series of dilators (8-, 12-, and 14-French) are passed over the guidewire, and an 18-French peel-away sheath is placed. A double-cuff, swan-neck, Tenckhoff peritoneal dialysis catheter is introduced over the stiff glide-wire into the peritoneal cavity. The coiled intraperitoneal portion is placed in the lower intraabdominal area. The inner cuff is pushed into the muscle before removing the peel-away sheath. A tunnel is created with an exit site located distal, lateral, and below the initial incision with the outer cuff buried in the subcutaneous tissue. A final fluoroscopic

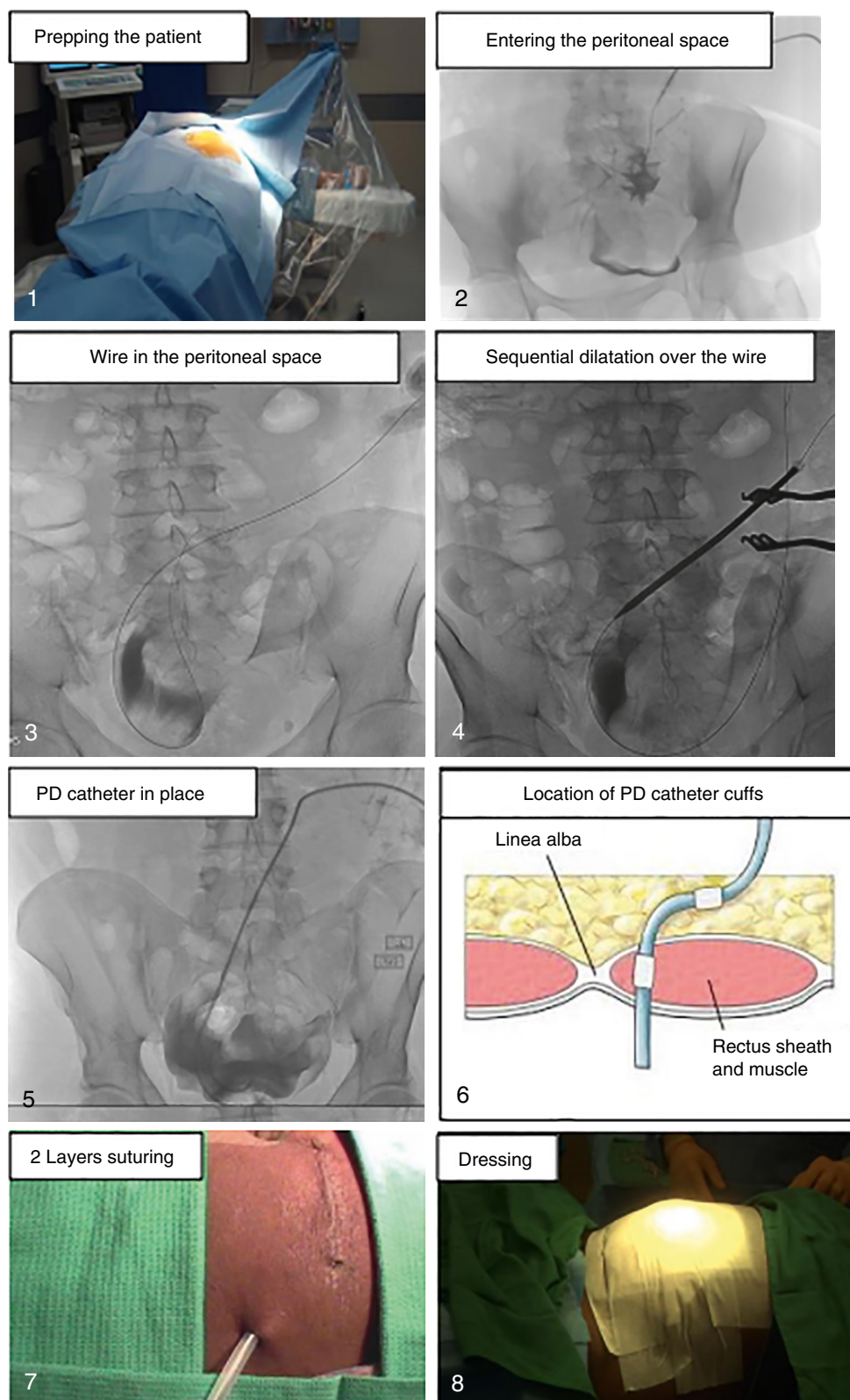


Fig. 67.39 1. Prepping the patient. 2. Entering the peritoneal space confirmed by free flow of contrast injection. 3. Wire in the peritoneal space. 4. Sequential dilatations. 5. Appropriate position of the PD catheter. 6. Location of the two cuffs. 7. Closing the incision. 8. Final dressing.

imaging is performed to verify placement of the Tenckhoff catheter (Fig. 67.39). Inflow and outflow of the PD catheter is tested with 500 to 1000 mL of normal saline. The PD catheter is flushed with 10 to 15 mL of heparin.

The subcutaneous tissue and skin are sutured, and the site is dressed²²⁸ (see https://cdn.ymaws.com/www.asdin.org/resource/resmgr/primer/IN_Primer_Part_2_Final_2020.pdf).

BEDSIDE PERITONEAL DIALYSIS INSERTION: ITS USE IN LOW-INCOME COUNTRIES AND DURING COVID-19

PD catheter insertion at the bedside holds many advantages over other techniques in specific circumstances, and knowledge of this technique by nephrologists has the potential to significantly improve PD uptake. In clinical practice, most PD catheters are placed by surgeons. As such, these procedures are delayed due to the limited surgical operating room time. Hence, bedside percutaneous insertion allows for rapid initiation of dialysis, especially for PD in acute kidney injury (AKI) or urgent-start dialysis in late-presentation ESKD.

In fact, studies in CKD from the United States, Canada, and Malaysia have shown that initiation of a percutaneous PD catheter insertion service significantly increased PD uptake, presumed due to accessibility to timely catheter insertion without the need for hemodialysis bridging.²⁵⁸⁻²⁶⁰ Other advantages of the percutaneous approach are that relatively small incisions result in reduced leak and infection rates, as well the ability to insert the catheter without the need for general anesthesia, which is especially useful in the face of hyperkalemia.

The modified Seldinger technique with a peel-away sheath introducer can be performed under local, regional, or general anesthesia. Absolute contraindications for this approach are limited to a previous midline incision laparotomy, those with a high risk of intraabdominal adhesions, and significant obesity with a skin-to-peritoneum distance >5.5 cm. Many centers perform this procedure routinely in patients who have had previous uncomplicated laparoscopic cholecystectomy, appendectomy, and Pfannenstiel incision caesarean section.

Within this context, there has been an increased interest in the use of PD for AKI with the “Saving Young Lives Program” in low-resource countries, an approach used in high-income countries during the COVID-19 pandemic (further discussed in [Chapter 73](#)). Both situations relied almost solely on percutaneous catheter insertion as patients couldn’t be moved from the critical care environment or needed the catheters to be inserted before operating room space became available. Excellent outcomes have been achieved even with the use of large fill volumes from the outset with minimal leak. It is thought that the dilatation of the tract using the Seldinger technique creates a tight seal around the catheter. This is not as easy to achieve with the surgical incision, which requires a purse-string suture technique to prevent leakage ([Fig. 67.40A and 67.40B](#)). The Saving Young Lives Program, which trains clinicians in low-resource settings to insert PD catheters for AKI, has trained more than 700 people in 49 countries using a model created with a pork belly suspended over a packing crate. This offers a low-cost, highly realistic option to train nephrologists in programs where PD numbers are small. Other options for training are offered through the International Society of Nephrology interventional nephrology fellowships further increasing competence; however, there is still a large gap in training for a skill that should arguably be learned by all nephrology trainees.

COMPLICATIONS OF PERITONEAL DIALYSIS CATHETERS

Several complications encountered after PD catheter placement can be categorized into mechanical and infectious groups.²⁶¹⁻²⁶³ Mechanical complications are related to either catheter flow dysfunction or catheter leakage²⁵³ (see https://cdn.ymaws.com/www.asdin.org/resource/resmgr/primer/IN_Primer_Part_2_Final_2020.pdf).

PERITONEAL DIALYSIS CATHETER FLOW DYSFUNCTION

- Extrinsic compression of catheter tip:** The main mechanism of this complication is the obstruction of the catheter side holes by distended bladder or colon (constipation).
- Luminal obstruction:** the key mechanism of this complication is tube kink or fibrin/blood clot that creates a two-way luminal obstruction ([Fig. 67.41A and 67.41B](#)). This complication is treated by either irrigating the catheter using saline or instilling tPA to dissolve the fibrin clot. Sometimes, the use of wires under fluoroscopic guidance is indicated. In rare cases, PD catheter requires exchange.
- Poor positioning and migration:** This complication results from the catheter resiliency memory forces that lead to the unusual position of the PD tip causing poor flow (see [Fig. 67.41C](#)). In most cases, fluoroscopic repositioning is attempted with limited success and subsequently the catheter is exchanged.
- Tissue attachment and entrapment:** In this scenario, the catheter is entrapped within the omentum or due to adhesions creating one-way or two-way poor flow (see [Fig. 67.41D](#)). These complications usually require laparoscopic intervention.

PERITONEAL DIALYSIS CATHETER LEAKAGE

- Pericatheter leak:** This complication is mostly seen in urgent-start PD. Therefore supine small-volume dialysate exchanges and decreased physical activity are recommended. In rare cases, this complication may require switching to hemodialysis temporarily.
- Abdominal wall hernia:** The mechanism of this complication is related to the increased intraabdominal pressure due to PD fluids. Obesity and steroid use are risk factors ([Fig. 67.42A](#)). In most cases, it requires surgical repair.
- Hydrothorax (pleuroperitoneal connection):** The mechanism is related to the movement of the dialysate from the peritoneal space into the pleural space through defects present within the diaphragm (see [Fig. 67.42B](#)). Treatment options include low-volume dialysate, pleurodesis, or pleuroscopic repair.

INFECTIOUS COMPLICATIONS OF PERITONEAL DIALYSIS CATHETER

Infectious complications are the most common cause for the loss of the PD catheter and switching to hemodialysis.^{261,262,264}



Fig. 67.40 Bedside placement of tunneled peritoneal dialysis catheter. (A) 1. A 2- to 4-cm horizontal or vertical incision is made in the skin 2 cm below the umbilicus. 2. Blunt dissect down to the linea alba. 3. Grab the rectus sheath with the artery forceps and lift the abdominal wall upwards. 4. Prepare guidewire and fill syringe with saline and attach the 16-gauge intravenous cannula. 5. The guidewire is passed through the plastic cannula and should feed easily. 6. The plastic cannula is removed and the peel-away introducer is fed over the guidewire. 7. Insert the catheter through the peel-away sheath until up to the first cuff. 8 and 9. Break the sheath and pull it apart. Bury the deep cuff in the subcutaneous tissue.

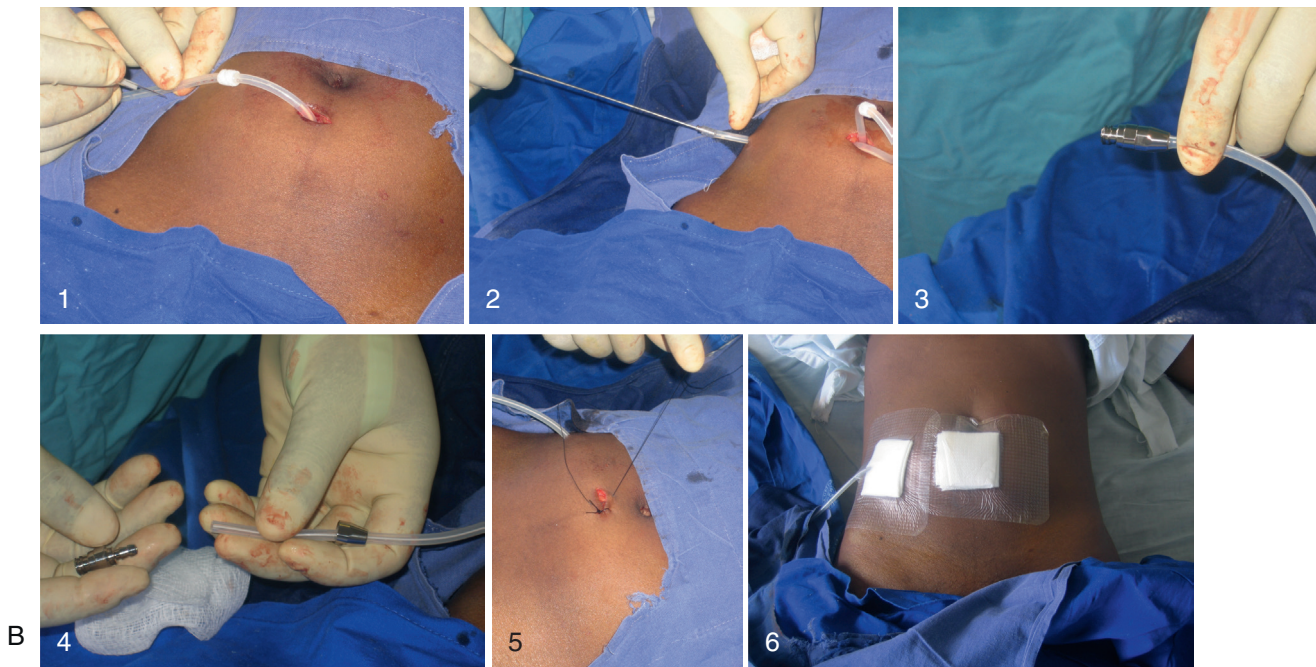


Fig. 67.40—cont'd (B) 1 and 2. Create the exit site and tunnel the catheter. 3 and 4. Attach the connector. 5 and 6. Close the incision and apply dry dressing.

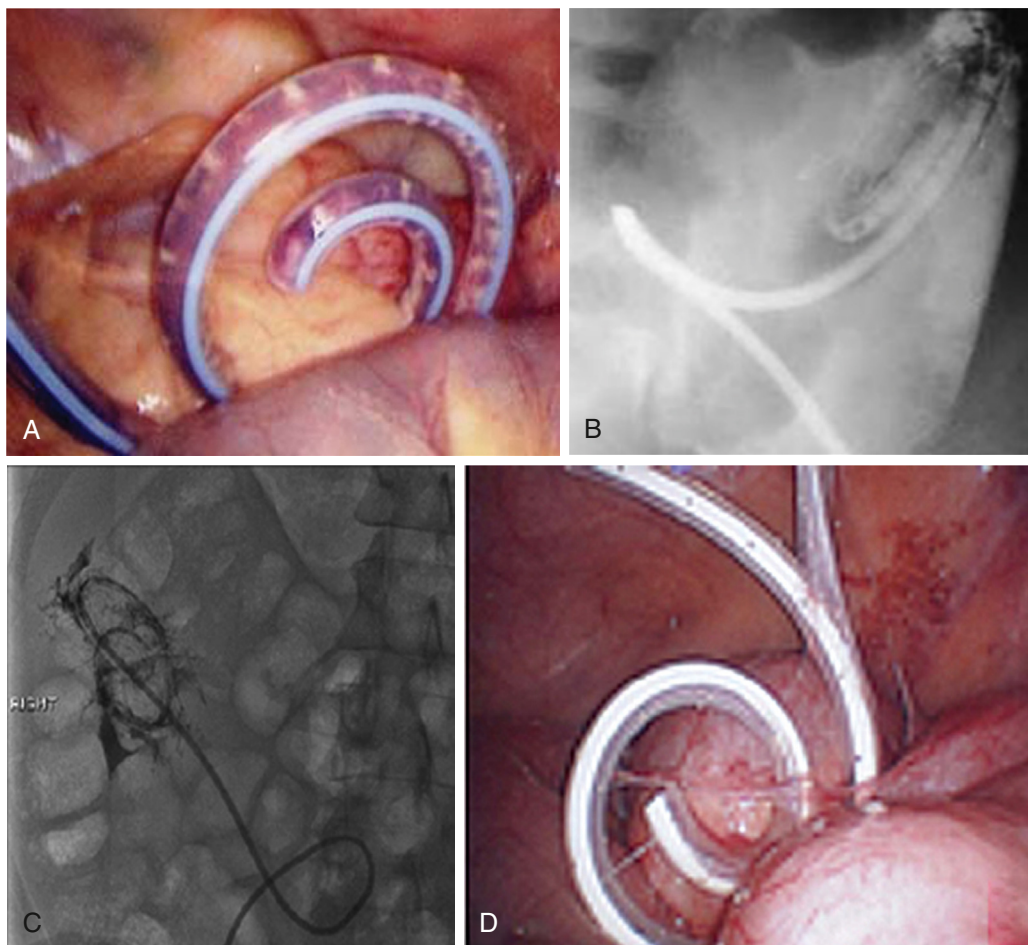


Fig. 67.41 (A) Fibrin and blood clot of the catheter. (B) Catheter kink. (C) Catheter migration. (D) Catheter entrapment. (From American Society of Diagnostic Interventional Nephrology in Primer. https://cdn.ymaws.com/www.asdin.org/resource/resmgr/primer/IN_Primer_Part_2_Final_2020.pdf)

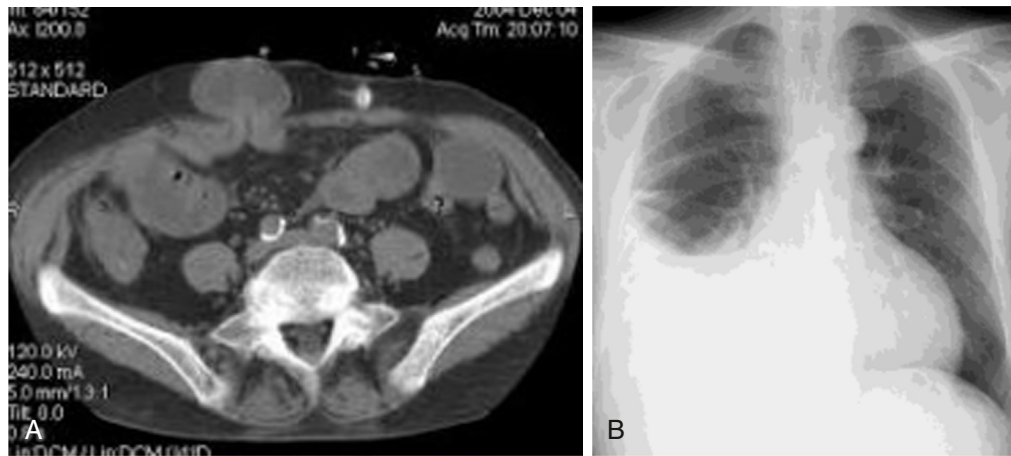


Fig. 67.42 (A) Ventral hernia. (B) Hydrothorax resulting from pleuroperitoneal connection.



Fig. 67.43 (A) Exit site infection. (B) Tunnel infection. (C) Cuff extrusion. (D) Macroscopic appearance of acute peritonitis. (Courtesy John Crabtree and Kaiser Belflower.)

- a. Exit site infection: This is usually related to poor exit site care. It requires drainage culture and antibiotic initiation (Fig. 67.43A). Tunnel ultrasound may be needed to rule out any tunnel involvement.²⁶⁵
- b. Tunnel infection: This may arise from exit site infection and necessitates tunnel ultrasound, drainage culture, and antibiotic initiation with *S. aureus* coverage (see Fig. 67.43B). In most cases, these catheters are exchanged if no response to antibiotic is encountered.^{261,262,264}
- c. Superficial cuff extrusion: This complication occurs when the superficial cuff rubs against the exit site and gets infected (see Fig. 67.43 C). Most cases are treated by exit site care, cuff shaving, and rarely catheter exchange.



Fig. 67.44 Effluent peritoneal dialysis color as a diagnostic tool.

d. Acute peritonitis: This complication is the main cause of switching to hemodialysis. Clinically, the main manifestations include abdominal pain and fever, accompanied with cloudy PD fluids (see Fig. 67.43D). The change of PD effluent color may indicate intraperitoneal or extra-peritoneal process; as such, the color change of PD effluent can be used as a diagnostic tool²⁶⁶ (Fig. 67.44).

Details of diagnosis and management of peritonitis are discussed in Chapter 63.

INDICATIONS FOR PERITONEAL DIALYSIS CATHETER REMOVAL

The key indications for PD catheter removal include refractory peritonitis, relapsing peritonitis, fungal peritonitis, and refractory exit site and tunnel infections. Additionally, removing PD catheter should be considered when peritonitis is not responding to therapy such as multiple enteric organisms and mycobacterial peritonitis.²⁶⁴

PERCUTANEOUS KIDNEY BIOPSY

A percutaneous kidney biopsy is an important procedure in the diagnosis of acute and chronic kidney disease. Kidney biopsy is discussed further in Chapter 25. The results of a kidney biopsy are helpful in guiding medical therapy and providing a prognosis. The goal of a kidney biopsy should be to maximize the yield of adequate tissue while minimizing the risk of complications. Percutaneous kidney biopsies have evolved from a blind procedure to a real-time ultrasound-guided needle biopsy. At some institutions, radiologists perform kidney biopsies under ultrasound or CT guidance. Most commonly, kidney biopsy is performed using real-time ultrasonography guidance along with the use of an automatic biopsy gun, an approach that minimizes procedure complications and provides a high yield of adequate tissue for pathologic diagnosis (Table 67.13).²⁶⁷⁻²⁷⁴ Cozens and colleagues²⁷² retrospectively compared a 15-gauge Tru-cut kidney biopsy with ultrasound localization and marking to use of an 18-gauge spring-loaded gun kidney biopsy under real-time ultrasound guidance. They reported a 79% yield of adequate kidney tissue with the blind technique (15 gauge), as compared with 93% with

Table 67.13 Adequacy of Kidney Tissue Retrieval and Complications by Real-Time Ultrasound-Guided Percutaneous Renal Biopsy

Reference	No. of Biopsies	Adequate Tissue	Major Complications ^a
Dowd, 1991 ²⁶⁷	23	95.5%	<0.5%
Doyle, 1994 ²⁶⁸	86	99%	0.8%
Hergesell, 1998 ²⁶⁹	1090	98.8%	<0.5%
Donovan, 1991 ²⁷⁰	192	97.8%	<1%
Burstein, 1993 ²⁷¹	200	97.5%	5.6%
Cozens, 1992 ²⁷²		93%	N/A
Marwah, 1996 ²⁷³	394		6.6%
Maya, 2007 ²⁷⁴	65	100%	0%

real-time ultrasound guidance (18 gauge).²⁷² Similarly, two other studies reported a higher mean number of glomeruli from biopsies obtained under real-time ultrasound, as compared with those performed blindly^{271,273} (see https://cdn.ymaws.com/www.asdin.org/resource/resmgr/primer/IN_Primer_Part_3_Final_Aug31.pdf).

Major complications of kidney biopsies including gross hematuria or retroperitoneal hematoma requiring blood transfusion, invasive procedure, or surgical intervention have been reported in <1% of biopsies in some series and 5% to 6% in others (see Table 67.13). The likelihood of major complications was not associated with patient age, blood pressure, or serum creatinine in one large series.²⁷³ Among those patients with major complications, the time interval from biopsy to diagnosis of the complication was ≤4 hours in 52%, ≤8 hours in 79%, and ≤12 hours in 100% of patients.²⁷³ A prospective study of 100 consecutive patients showed that outpatient, real-time, ultrasound-guided percutaneous kidney biopsy is safe and minimizes the need

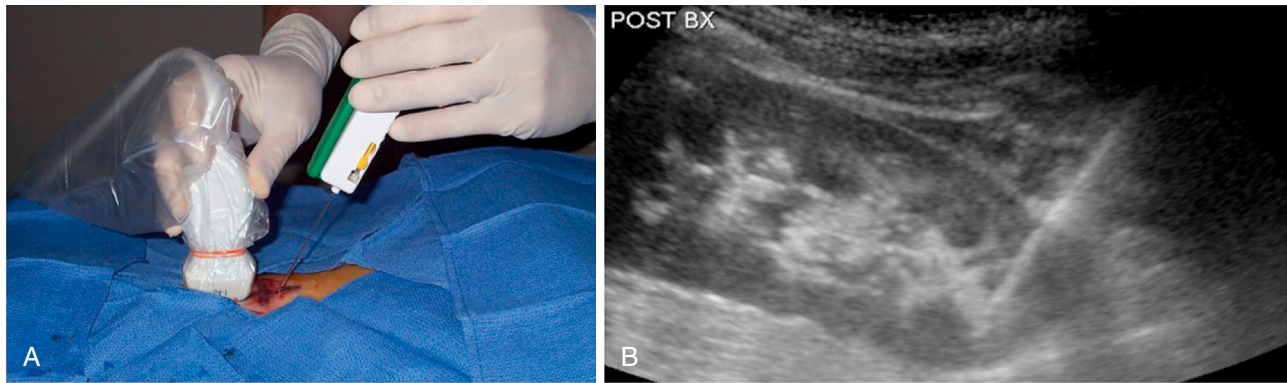


Fig. 67.45 (A and B) Use of real-time ultrasound guidance to perform kidney biopsy.

for postbiopsy hospitalization.²⁷³ Minor complications including transient gross hematuria or perinephric hematoma not requiring transfusion or intervention occurred after 6.6% of biopsies in one series.²⁷³ Either ultrasound or computed tomography can be used to diagnose perinephric hematomas.²⁷⁵ Most hematomas resolve spontaneously within a few weeks with no significant sequelae. Major bleeding complications that do not resolve with conservative measures require further intervention. In clinical practice, these complications are often treated endovascularly by performing selective renal arteriogram and embolizing the bleeding arteriole. Surgical intervention with nephrectomy is rarely needed. A review article reported major complications in only 0.3% of cases, with death in <0.1% among 9595 percutaneous kidney biopsies performed over the past 50 years.²⁷⁶

For patients at high risk of bleeding complications or liver disease with coagulopathy in whom a kidney biopsy is indicated, a transjugular kidney biopsy may be performed by an interventional radiologist.²⁷⁷

A bleeding disorder is an absolute contraindication to performing a percutaneous kidney biopsy. However, if it can be corrected medically, and if the potential benefit of doing a biopsy outweighs the potential risk, the biopsy can still be performed. Relative contraindications to kidney biopsy include a solitary kidney, pyelonephritis, perinephric abscess, uncontrolled hypertension, hydronephrosis, polycystic kidney disease, severe anemia, pregnancy, renal masses, and renal artery aneurysms.

PERCUTANEOUS KIDNEY BIOPSY UNDER REAL-TIME ULTRASOUND GUIDANCE

While the patient is in a prone position, an initial ultrasound examination is performed to confirm the presence of two kidneys. The lower pole of the left kidney is preferred for right-handed operators. Using a sterile technique, the skin and subcutaneous tissue are anesthetized with 1% lidocaine. Under real-time ultrasound guidance, a biopsy needle gun is advanced up to the capsule of the kidney (Fig. 67.45A and 67.45B). Next, the spring-loaded gun is activated while the patient holds a breath. The gun is retrieved, and the specimen is placed in a container with media. Two to three biopsy samples are taken in one setting to provide enough tissue for light microscopy, immunofluorescence, and

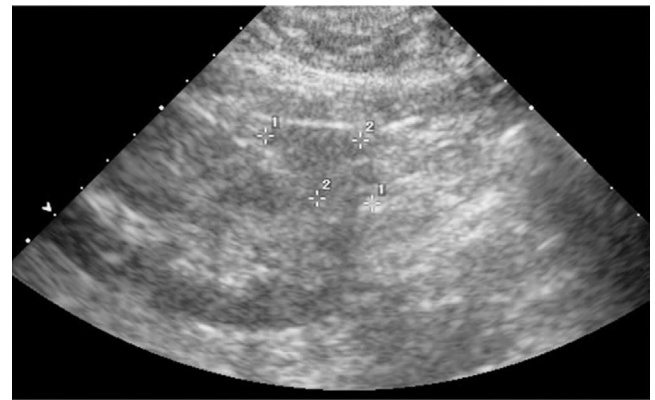


Fig. 67.46 Perinephric hematoma seen after kidney biopsy.

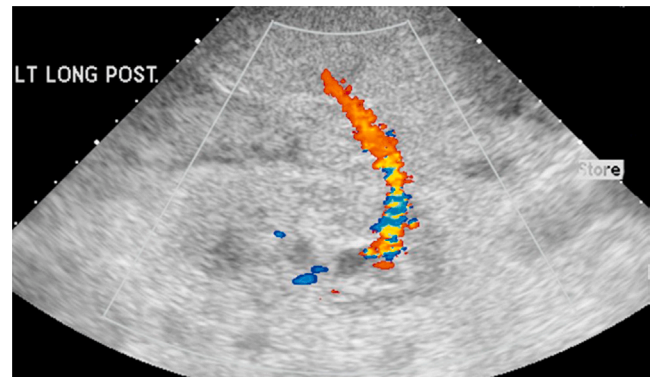


Fig. 67.47 Color Doppler ultrasound surveillance after kidney biopsy. Abnormal color Doppler ultrasound that shows the needle track that extends out of the kidney into the adjacent hematoma.

electron microscopy studies. After the biopsy is completed, a second-look ultrasound examination is performed to assess for perinephric hematomas (Fig. 67.46). A color Doppler ultrasound post biopsy surveillance imaging examination is helpful to localize any active bleeding (Fig. 67.47).

Most patients are typically observed for at least 6 to 8 hours after a kidney biopsy, with serial measurements of vital signs and hemoglobin to monitor for any major hemorrhage. However, some evidence suggests that if the immediate postbiopsy kidney ultrasound is negative for active bleeding, it is extremely unlikely that a late major hemorrhagic complication can occur (negative

predictive value, 100%). As such, those patients can be discharged home safely within 2 to 4 hours of the procedure.²⁷⁸

RADIATION AND PERSONAL SAFETY

The understanding of basic radiation safety is important to protect the patient, physician, and staff involved in the care of the patient in interventional nephrology. Unnecessary radiation exposure is harmful and can be easily prevented. The Food and Drug Administration (FDA) oversees the rules and regulations for use of x-ray equipment. The Occupational Safety and Health Administration (OSHA) regulates the radiation exposure of workers. Each state has its own regulatory office to ensure that workers do not exceed a predetermined radiation dose.

Exposure is the amount of ionizing radiation reaching a subject and is measured in units of Roentgen. The amount of energy absorbed by a material when it is exposed to ionizing radiation is measured in Rads. The absorbed dose is always lower than the exposure because the tissue does not absorb all the energy from the radiation. An absorbed dose equivalent is used to relate the amount of biological damage and is measured in Rem. The OSHA occupational dose limit for the whole body is 1.25 Rem/quarter; for the extremities it is 18.75 Rem/quarter. A dosimeter must be always worn on the outside of the lead apron, and the absorbed dose is measured monthly. To protect against radiation, the interventionist should minimize the time of exposure to radiation, minimize the use of magnification imaging, properly use collimators and field filters, maximize the distance between the source of radiation and personnel involved with the procedure, minimize the use of cineangiography and continuous fluoroscopy, and use proper shielding including lead aprons, thyroid collars, leaded glasses, and lead shields.

Knowledge of these facts and application of appropriate safeguards is particularly important in hemodialysis access procedures, especially those involving vascular access interventions in the upper extremity. The operators' proximity to the x-ray tube and difficulty in shielding increases their radiation exposure.

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